DIAGNOSING AND FIXING POLARIZATION ATTRIBUTION TO GROUPS IN SUBJECTIVE ANNOTATIONS

Anonymous authors
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ABSTRACT

Current representations of polarization focus exclusively on distinguishing minor disagreements from systematic differences in opinion between annotator groups. In this study we develop a new framework which not only detects, but also quantifies the degree to which any annotator subgroup causes these systematic differences. We first identify fundamental issues and demonstrate how they limit existing approaches: 1) the presence of "inherent" polarization that cannot be attributed to any known or latent groups, and 2) opposing polarization effects canceling each other out in aggregated annotations. We then introduce a new metric that attributes polarization to any annotator group while avoiding the above limitations and the persistent problem of group-size imbalances, releasing a PyPi-installable implementation. Our metric surpasses both traditional similarity metrics and prior approaches on detecting subgroup polarization, while also uncovering previously latent polarization in overlooked minority and fine-grained ideological groups. Finally, we explore the polarization patterns of persona-prompted LLM-as-a-judge models, finding that they are disconected from human polarization patterns, even when explicitly instructed to condition responses on persona characteristics.

1 INTRODUCTION

Annotations are essential for subjective tasks, such as content moderation, toxicity, and Hate Speech (HS) detection, which are especially important for protecting minority groups United Nations (2025); Dioum (2025). However, in practice, analysis rests on inter-annotator agreement or majority-vote Ivey et al. (2025); Kralj Novak et al. (2022); van der Velden et al. (2025), which may lead to biased and misconfigured systems (e.g., toxicity models disproportionally marking African American (AA) speech as "toxic" Sap et al. (2019)), or discard *fundamental* disagreements between minority groups and the majority Quaranto (2022), (see the example in Fig. 1).[1]

One alternative is using *polarization*, which correlates better with human judgments and is much less sensitive to different group sizes Pavlopoulos & Likas (2023; 2024). While we can detect polarization, there is little work on how to detect whether it is *actually caused by marginalized groups*. Our work thus tries to answer the following **Main Research Question (MRQ): How can we build a robust representation of intergroup polarization?** We answer this question through the following sub-questions:

RQ 1: How much of the observed polarization is attributable to annotator subgroups? While prior work assumed that all polarization can be explained and attributed to some annotator characteristic, through the exhaustive creation of theoretical annotator groups (§4) our study discovers the existence of Inherent Polarization (InPol), which can not be attributed to *any possible group* of annotators—known or unknown (Finding 1). We also discover that polarization across items has a *direction*, preventing us from aggregating annotations on the dataset level to circumvent annotation sparsity (Finding 2), which we later show to limit existing statistical approaches.

RQ 2: How can we quantify this attribution to each possible subgroup? In order to solve these issues we introduce a new metric "*apunim*", which *quantitatively* attributes polarization to specific annotator sociodemographic and ideological characteristics (referred to as Personal Characteristics

[1] The simulated experiments in this study were implemented by generating random annotations following the normal distribution with differing means and Standard Deviation (SD)—see App. G.

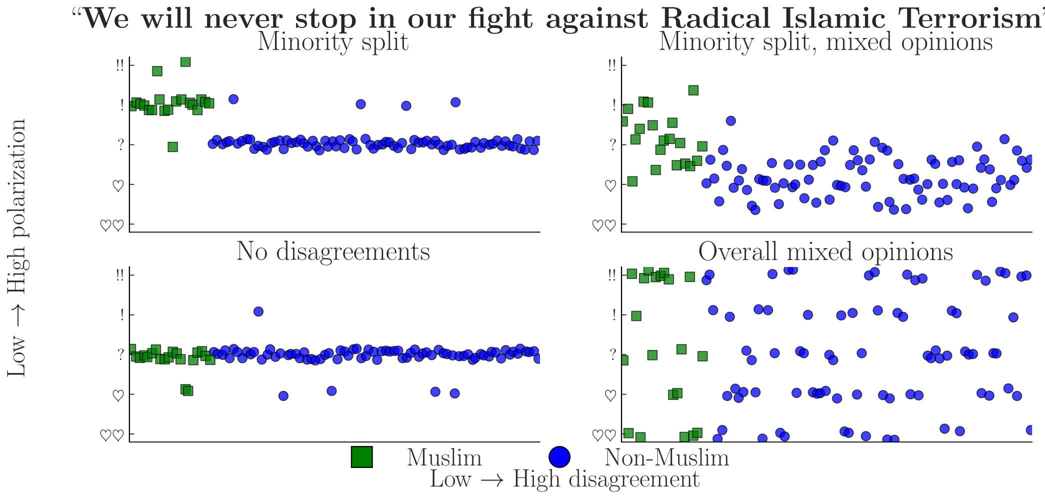

Figure 1: Simulated experiment exploring four scenarios of a HS detection task applied to a controversial statement by D. Trump (Jan. 2019). The variance in annotations across different annotators may cause traditional agreement metrics to ignore minority opinions, especially when opinions are mixed (top-right).

(PCs) in this study) and outperforms prior approaches in scenarios with realistic annotation sparsity and minority group splits (Finding 3).

RQ 3: Which polarization patterns are detectable using our new representation? We apply our metric to four Natural Language Processing (NLP) datasets focused on Toxicity and HS detection (§6), confirming prior work showing that annotator gender and ethnicity significantly impact annotation, while discovering that ideological attributes and personal experiences may play a major role in explaining polarization (e.g., the more religious the annotators, the more polarized they become). These possibilities have not adequately explored by relevant literature (Finding 4).

RQ 4: Which polarization patterns are shared between human and LLM annotators? We conduct an annotation study on the same datasets using six open-source models in order to compare polarization patterns between them and humans (§6.2). We find that persona-prompted LLMs consistently fail to produce any subgroup polarization patterns, confirming that persona prompting does not consistently influence LLM outputs (Finding 5).

Our work makes the following contributions: (1) The identification of severe issues present in subjective datasets and prior methodologies (2) A new polarization attribution metric that is robust to small annotator counts and minority group splits with a PyPi-installable implementation (3) The detection of new promising directions (ideological and ordinal groups) for analysis of subjective NLP datasets (4) An exploration of LLM polarization patterns when using persona prompting. Our study also underscores the need for datasets with high annotator counts per item, instead of high annotator counts in general—see App. I for a more extensive discussion, empirical findings, and an intuition on how polarization attribution could be used for filtering subjective datasets.

Content Warning: Controversial and potentially harmful speech used for demonstration.

2 Background and Related Work

2.1 Agreement

Popular inter-annotator agreement metrics such as Cohen's kappa, Krippendorff's alpha, and Fleiss's kappa aim to distinguish genuine disagreement from random disagreement through statistical chance-correction. However, these metrics are often difficult to apply to intergroup agreement without extensions or restrictive assumptions (e.g., no missing labels or equal numbers of annotations). They have also been criticized for failing to account for differences arising from social, cultural, and demographic backgrounds (Feinstein & Cicchetti, 1990; Byrt et al., 1993; van der Velden et al., 2025; Checco et al., 2017), producing results that are difficult to compare across studies, being sensitive to class imbalance (Feinstein & Cicchetti, 1990), and relying on questionable assumptions about chance agreement Checco et al. (2017). While recent work has addressed some of these limitations with methods that improve applicability and robustness van der Velden et al. (2025); Wong et al. (2021); Ivey et al. (2025), they are much more complex and less intuitive compared to their traditional counterparts, indicating that agreement may not be the correct tool for analysis of minority viewpoints.

2.2 Polarization

There are multiple competing formulations for polarization. One of these approaches Akhtar et al. (2019) utilizes χ^2 to estimate in-group vs out-group variance, but still requires traditional agreement metrics for analysis. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric Checco et al. (2017), or non-parametric Pavlopoulos & Likas (2024); Mignemi et al. (2024) approaches to the annotation histograms.

For our work, we use a non-parametric polarization metric, normalized Distance From Unimodality (nDFU), which has been shown to correlate better with human judgment of disagreement than established metrics (Pavlopoulos & Likas, 2024). nDFU (Eq. 1) takes values in the range $[0, 1]$, where values close to 0 indicate no polarization (i.e., opinions are concentrated around a single mode). Intuitively, nDFU measures how prominent secondary peaks are compared to the main peak (see Fig. 2, left for an example).

$$\mathrm{nDFU}(X) = \frac{\max\limits_{i \neq m} \left(f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right)}{f_m}, m = \arg\max_i f_i \tag{1}$$

where, X denotes the set of annotations, f_i the relative frequency of the i-th annotation (e.g., the frequency of 2 in a scale of 1–5), and m the histogram peak.

2.3 Attributing Polarization to Groups

While many studies study the effect of subgroups on annotation agreement Goyal et al. (2022); Binns et al. (2017); Giorgi et al. (2025); Al Kuwatly et al. (2020); Sachdeva et al. (2022), attributing *polarization* is a relatively new concept. Under the name *aposteriori unimodality* Pavlopoulos & Likas (2024), the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then it is explained by those groups. Though promising, this mechanism operates on the assumption that *all* polarization can be explained by a single split, which we disprove prove to be impossible in many cases (§4). Additionally, getting polarization estimates from few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarizing) instead of a quantifiable value.[2]

3 Datasets

Dataset	#Com	#Ann	#Dims	Task
Kumar	2998	5	10	Tox
Sap	585	5–6	3	HS
DICES-350	350	123	4	HS
DICES-990	990	70–76	4	HS

Table 1: Datasets used in this study. *#Ann* refers to the number of annotations per item (see App. B), and *#Dims* to PC dimensions (e.g., age, gender). Tasks are either Toxicity (Tox) or HS.

We use four datasets from current NLP research that include PC information at the level of individual annotators; specifically, the “Kumar” Kumar et al. (2021) and “Sap” Sap et al. (2022) datasets, which

[2] For example, when presented with noisy estimates of a metric in the $[0, 1]$ range, we would ideally want to distinguish between cases where attribution is close to $.001$, and cases where attribution is $.49$.

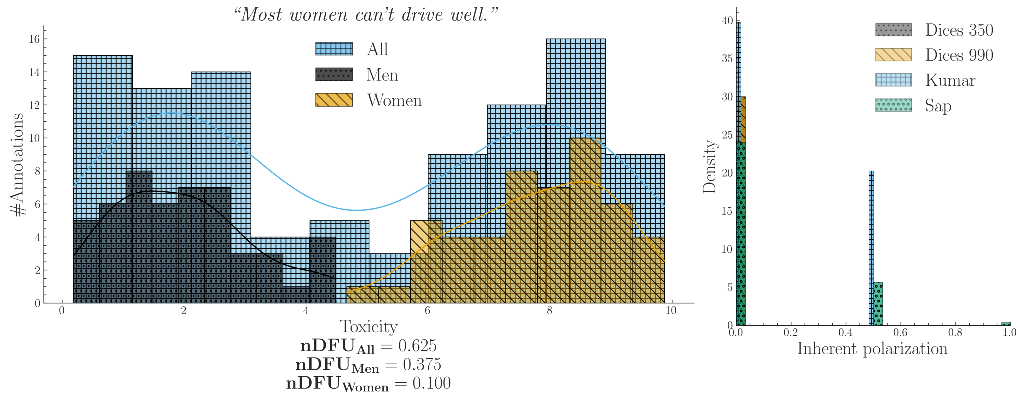

Figure 2: **Left:** Simulated experiment describing a polarizing comment where male and female annotators agree between themselves, but disagree with the opposite gender. **Right:** InPol across our datasets (§3). Contrary to assumptions made in prior work, InPol is commonly non-zero and can even reach $nDFU = 1$ in datasets with low annotator counts.

are concerned with Toxicity and HS detection respectively. We also examine the DICES-350 and 990 datasets Aroyo et al. (2023) which focus on LLM response safety.[3]

A brief overview of dataset characteristics is provided in Table 1. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated items, as well as 10 distinct PC dimensions.[4] In contrast, the DICES datasets contain fewer annotated items but involve more than an order of magnitude more annotators per item than the other datasets. More details on the annotations, including overall levels of observed polarization, can be found in App. B.

4 Can We Attribute All Polarization to Annotator Groups?

4.1 Problem formulation

Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated items.[5] Each item c is assigned multiple annotators, which creates the annotation multi-set $A(c) = \{(a_1, g_1), (a_2, g_2), \ldots\}$, where a_i is a single annotation (e.g., 2 in a LIKERT scale of 1–5), and g_i a group (e.g., men). As an example, the annotations for a item in a sentiment analysis task where we group the annotators by gender, would be formulated as:

$$A(\text{“Could be better, could be worse”}) = \{\ \ (+, F), (-, M), (+, F), \ldots\ \ \}$$

Intuitively, a group partially explains the polarization of a item c when the annotations of that group $A(c \mid g) = \{a(c, g') \in A(c) \mid g' = g\}$ exhibit higher polarization compared to the full set of annotations $A(c)$. Consider the case where a misogynistic comment is annotated for toxicity by male and female annotators shown in Fig. 2, left. The annotations are generally polarized ($nDFU_{all} = .62$); but both female ($nDFU_{women} = .10$) and male ($nDFU_{men} = .37$) annotators exhibit low polarization between their respective groups. This pattern suggests that the overall polarization is partially caused by substantive disagreements between male and female annotators.[6]

[3]Specifically, we use the "Breadth of Posts" dataset with the "*toyou*" labels for Sap. For DICES-350 we choose the *Q3_bias_targeting_inherited_attributes* label, while for DICES-990 the *Q3_bias_incites_hatred* forgoing the *Q3_bias_overall* labels which are aggregated scores. The choice of labels transforms DICES into a HS task with the most coherence since the two datasets do not share labels.

[4]We sample 3,000 items in this study due to computational constraints (App. D). Our results remain consistent, for both different and differently sized-samples (App. H).

[5]The items could be of any modality (e.g., images, videos, survey questions).

[6]Since polarization persists among male annotators, another subgroup of men may also contribute to it.

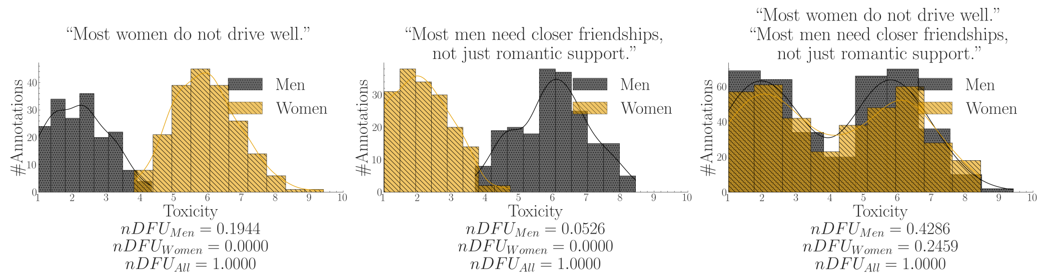

Figure 3: A synthetic experiment simulating a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments, we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions, obscuring the source of the polarization (polarization explained by gender, but not shown).

4.2 Inherent polarization

What Is It? Inherent Polarization (InPol) refers to disagreement that cannot be attributed to *any specific* annotator split representing arbitrary groups. Prior work Pavlopoulos & Likas (2024) attributed polarization to groups only when polarization for each individual group became exactly zero (e.g. $nDFU(all) > nDFU(men) = nDFU(women) = 0$). The presence of InPol would invalidate this theory, since there would be no possible group or combination of groups g *s.t.* $nDFU(g) = 0$. Formally, the InPol for a single item c is the minimum polarization exhibited by any arbitrary group (Eq. 2).

$$Pol_{inh}(c) = \min_{S \subseteq A(c),\, |S| \geq 3} \{\textbf{\textit{nDFU}}(S)\} \tag{2}$$

When considered exhaustively, these theoretical, random groups include both observed and latent groups (e.g., annotators may be less strict if they had lunch prior to annotation). Since nDFU operates only when a group has at least three annotators (Eq. 1), this yields $\sum_{k=3}^{n} \binom{n}{k}$ possible groups. Since enumerating all subsets is computationally intractable for large annotator counts, we use Monte Carlo sampling by drawing k random partitions with random sizes (Eq. 3) which serves as an upper bound since it considers a strict subset of Eq. 2.

$$Pol_{inh}^{MC}(c) = \min_{j=1}^{k} \{\textbf{\textit{nDFU}}(\tilde{r}_j(A(c)))\} \Rightarrow Pol_{inh}^{MC}(c) \geq Pol_{inh}(c) \forall c \tag{3}$$

where $\tilde{r}$ is an operator that selects a randomly sized, randomly selected subset.

Does It Exist? We compute the InPol for each item across the four datasets. For Sap and Kumar, we use the exhaustive formulation (Eq. 2), whereas for the two DICES datasets we rely on the Monte Carlo approximation with $k = 1000$ (Eq. 3). Fig. 2, right shows that in all DICES items there exists at least one group that fully accounts for the observed polarization.[7] Interestingly, this observation does not hold for the Kumar and Sap datasets, despite evaluating all possible annotator groups exhaustively. See App. B;Table 9 for a qualitative look through items with high InPol.

Finding 1. *Items may exhibit a degree of InPol that cannot be attributed to any known or latent annotator grouping given the annotators present in the dataset.*

Can We Remove It? Finding the reason behind InPol is difficult given the lack of datasets (see App. F for some hypotheses and further analysis). If we assume that InPol is caused by a lack of annotations per item, we could address the issue by aggregating annotations between items. However, Fig. 3 shows that two items polarized for the opposite reason may lead to a false negative. Therefore, any statistics must never be aggregated across items.[8]

[7] $Pol_{inh}^{MC}(c) = 0 \Rightarrow Pol_{inh}(c) = 0$ due to Eq. 3, as is the case of increased k in Eq. 3.

[8] A similar assumption was followed in early parametric approaches, which assumed different parameters for each item Checco et al. (2017).

Finding 2. *Items hold their own pol. direction which may compromise cross-item aggregation.*

5 How Do We Attribute the Rest?

5.1 Quantifying Attribution to Annotator Groups

Instead of using InPol, which represents the minimum possible polarization exhibited by any group, we calculate the *expected* (average). Therefore, we do not test whether the group is polarized against the whole; nor whether the group explains *some polarization* (as would be the case if we used Eq. 2), but rather whether the group explains polarization *more than other groups*. Specifically, we define apriori polarization for an item c according to Eq. 4, which is averaged across items for a dataset d (Eq. 5). Intuitively, a high apriori polarization score means that much of the polarization in the dataset is likely not driven by any single group.

$$Pol_{apr}(c) = \frac{1}{t}\sum_{i=1}^{t} \textit{nDFU}(\tilde{r}_i(A(c))) \quad (4) \qquad Pol_{apr}(d) = \frac{1}{|d|}\sum_{c \in d} Pol_{apr}(c) \quad (5)$$

where $\tilde{r}$ is the random partition operator (see Eq. 2). We explain why defining apriori polarization in this way is preferable to alternatives in App. E.

We compare this quantity with the average observed polarization of all items (Eq. 6), which we use to compare the polarization that is attributed to a single annotator group with the expected one (Eq. 5), yielding our normalized metric, *apunim* (Eq. 7).

$$Pol_{obs}(d, g) = \frac{1}{|d|}\sum_{c \in d} \textit{nDFU}(A(c \mid g)) \quad (6) \qquad apunim(d, g) = \frac{Pol_{apr}(d) - Pol_{obs}(d, g)}{1 - Pol_{apr}(d)} \quad (7)$$

Apunim measures the degree to which an annotator group contributes to overall polarization in a set of items. The *sign* of apunim carries information agreement-based attribution does not:

- $apunim(g) \approx 0$: The group g does not meaningfully disagree with the other annotators.
- $apunim(g) > 0$: The group g has fundamental disagreements with the other annotators.
- $apunim(g) < 0$: The annotators of group g are polarized between themselves.

We also include a perturbation-based p-value estimation, which tests whether the observed apunim value of a group g statistically surpasses that of random groups in App. C.

5.2 Comparison With Other Methods

We construct polarization distributions annotated by two groups. Each annotator is slightly biased to judge similarly with their group, although the group effect is different for each item. We model the disagreement between these groups using a parameter δ; higher δ signifies a larger divergence between the groups. An effect is considered detected if the statistic recovered using perturbations of a null distribution is significant. The complete experimental setup is detailed in App. G.

Fig. 4 shows the effectiveness of apunim, the aposteriori unimodality framework proposed by Pavlopoulos & Likas (2024), the χ^2 method presented in Akhtar et al. (2019), GMM clustering used in Checco et al. (2017), and a baseline using intergroup disagreement ($\Delta\alpha$ Krippendorff). Both the χ^2 and GMM methods generally fail at attributing polarization, which is most likely caused by opposing polarization effects (see Fig. 3) violating their assumptions on fixed mean differences between groups.[9] Intergroup disagreement remains competitive when many annotators are available, but underperform in cases where the examined group is a minority and when a few annotators are available per item (a common case among NLP datasets—see §3). Apunim in contrast is robust

[9]Indeed, re-running the experiment with a consistent mean-difference effect among the groups leads to exceptional performance by these methods—see App. H.1.

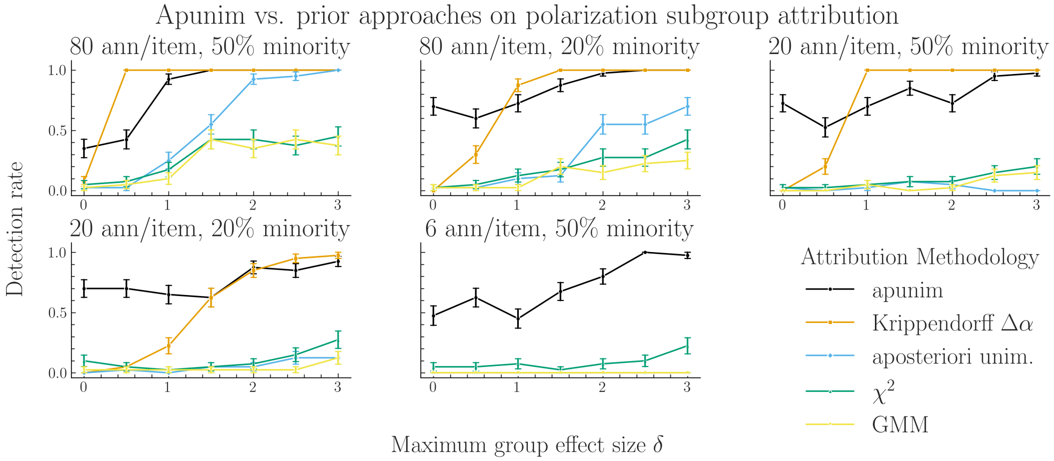

Figure 4: Polarization group attribution on randomly generated, synthetic distributions. Bars indicate standard error. For each experiment we assign a number of annotators per item, and split the annotator groups 50-50 or 80-20. Higher δ represents higher polarization between groups. Apunim is adept at attributing polarization to the correct group regardless of number of annotators per item, or how these annotators are split to different groups.

both to the number of annotators per item, opposing polarization effects, and minority groups, which validates our initial motivation.

Finding 3. *Apunim beats established and proposed methodologies in realistic conditions.*

6 Attributing Real-Life Polarization

6.1 Human Annotators

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided PC dimensions can explain observed polarization. Statistically significant results are presented in Table 2. See App. D for details on other parameters and execution times, and App. K for the full results. We note that the lack of rich annotation data prevents us from analyzing how different annotator groups interact (e.g., the relationship between LGBTQ+ and Conservative annotators) and thus our analysis is necessarily uni-dimensional.

*Polarization is consistently attributed to **ethnicity** across all datasets*, confirming prior work suggesting that it is a systematic source of annotator disagreement in these tasks Goyal et al. (2022).[10] Additionally, in both DICES-990 and Sap, ***women** are a consistent source of polarization* (positive values), while **men** disagree among themselves (negative values), which is consistent with previous findings (Wojatzki et al., 2018; Binns et al., 2017). Non-**transgender** annotators also showed the *highest degree of polarization* in the Kumar dataset in absolute terms (-0.788) suggesting that transgender individuals are a systematic source of disagreement.[11]

***Ideology and experiences** appear to be a more consistent source of annotator disagreement than inherent characteristics*. Specifically, the Kumar dataset, which tracks ideological attributes, shows polarization attribution linked primarily to personal experiences and beliefs, rather than to sociodemographic dimensions, although further datasets are required to generalize this finding. The divide between ideology/personal experiences and innate sociodemographic characteristics has not been adequately explored in current literature and should be further studied in future work. Furthermore, ***religion** demonstrates a clear relationship with polarization attribution*. According to Fig. 5, highly devout annotators exhibit higher levels of self-agreement compared with other annotators (posi-

[10] We note that polarization attribution does not have a consistent direction across datasets, which we believe to be caused by studies using different ethnic subgroups, allowing polarization detection on the level of PCs, but distorting subgroup attribution.

[11] Only the inverse effect is detectable due to annotator sparsity, hence the negative sign.

Group	Apunim	Support
DICES-350		
Ethnicity=AfricanAm	-0.037	7627
Ethnicity=Asian	0.045	6838
Education=Other	-0.023	1841
DICES-990		
Ethnicity=AfricanAm	0.099	5148
Ethnicity=Latino	-0.063	5877
Ethnicity=Other	0.034	21669
Ethnicity=White	-0.065	10166
Age=GenZ	0.047	12214
Age=Millennial	-0.007	34433
Age=GenX+	-0.028	12699
Education=College	0.006	52727
Education=High school	-0.050	6619
Gender=Man	0.036	28817
Gender=Woman	-0.027	29990
Sap		
Ethnicity=AfricanAm	0.042	4
Gender=Man	-0.091	1439
Gender=Woman	0.244	877

Group	Apunim	Support
Kumar		
A. Innate Characteristics		
Ethnicity=White	-0.029	6619
Education=College	-0.034	2508
SexOrient=Heterosexual	-0.050	5721
IsTrans=No	-0.079	1697
B. Experiences and Ideologies		
HasBeenTargeted=False	-0.048	6632
IsParent=No	-0.069	5514
IsParent=Yes	0.049	6216
ReligionImportant=No	-0.062	3864
ReligionImportant=Very	0.052	4062
SeenToxicity=False	0.035	3050
SeenToxicity=True	-0.077	6360

Table 2: Statistically significant ($p < 0.05$) apunim results across the four datasets. Gray rows show `Ethnicity` and `Gender` groups. “Support” refers to number of annotations. Consult App. K for the full results.

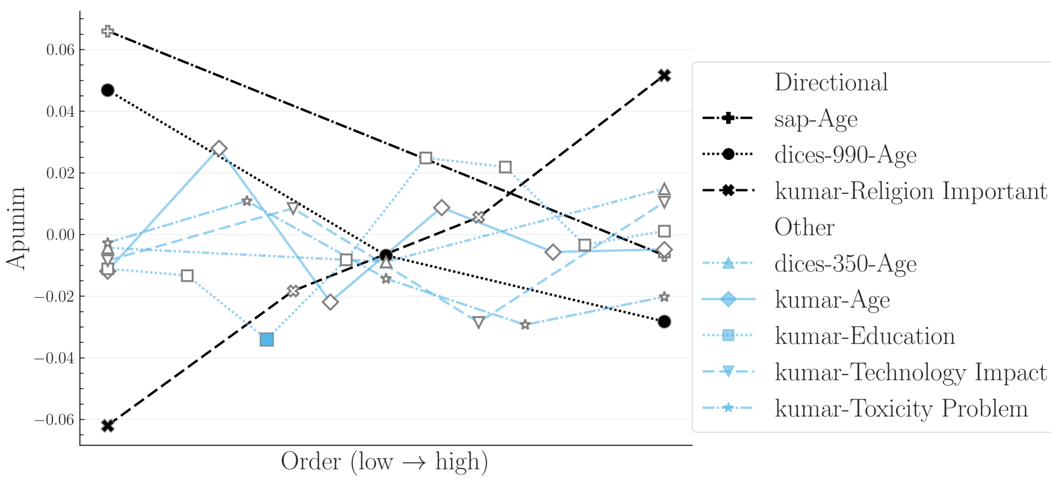

Figure 5: Polarization attribution trends in ordinal groups. The left side of the x-axis refers to young annotators, low education, low religiousness, and negativity towards the impacts of toxicity and technology. The full ordinal modes for each dimension and their individual labels can be found in App. K. We exclude groups with < 50 annotations, since they near-universally result in zero apunim values due to lack of data. Filled points represent statistical significant ($p < 0.05$) values.

tive apunim). Non-religious annotators show a pattern of increased internal disagreement (negative apunim), a trend that gradually reverses as religious devoutness increases (increasing apunim).[12]

[12]The statistical insignificance observed during the transition from irreligious to highly religious groups may represent genuine shifts rather than inconclusive results, since Eq. 8 checks whether $apunim \neq 0$, given that $\bar{apunim}_{\text{rand}}(d, g)$ will usually be close to zero. This phenomenon may also apply to other ordinal PCs, such as `Age` in both Sap and DICES-990, but finer-grained subgroups are needed to verify this.

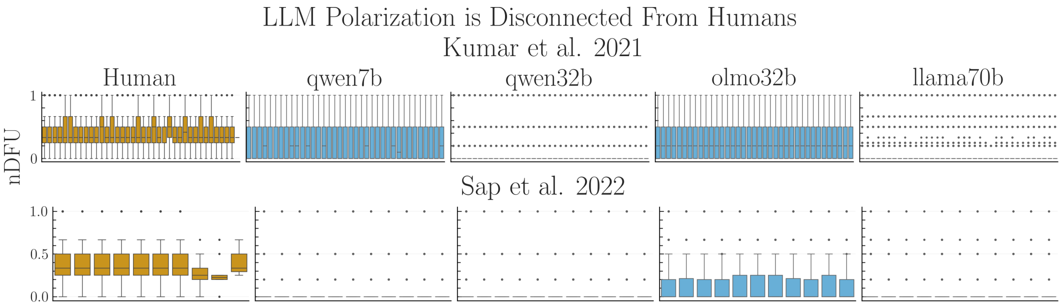

Figure 6: nDFU grouped by dataset (rows) and model family (columns) for each PC persona dimension in the primary annotation task (blue) using the default prompt, compared with human annotations on the same sample (yellow).

Finding 4. *Annotator ethnicity and gender are consistent causes of polarization, while the latter may be more pronounced in ideological attributes and personal experiences. Polarization also escalates as annotators become more religious.*

6.2 What Polarization Patterns Do LLM Annotators Exhibit?

Prior literature has shown that LLMs are not reliable annotators when using persona prompting; they tend to reproduce stereotypes (Lutz et al., 2025; Liu et al., 2024), are unreliable for minority groups (Lutz et al., 2025) and persona adherence has been shown to be volatile (Shu et al., 2024; Kovač et al., 2024). Unlike prior work that investigated model performance (Gupta et al., 2024) or human–LLM agreement (Dong et al., 2024; Gomez et al., 2024; Masud et al., 2024), we aim to discover whether attributable polarization exists regardless of faithfulness to the actual groups.

We investigate six open-source LLMs with six annotators per item supplied with personas sampled from our human dataset. We annotate a random sample of 300 comments for the Sap dataset and 1000 items for Kumar. Our results show *no statistically significant attribution effects* and very small *apunim* values for any of the groups in all models and datasets (see App. J; Table 17, 18). Using four adversarial prompts progressively emphasizing the importance of the LLMs' personas (App. L), since instruction prompts are known to influence the effect of personas (Weeber et al., 2026), also yields no results. Indeed, the adversarial prompts did not seem to push the *apunim* values to any particular direction, suggesting that instructing the model is an ineffective way of eliciting *any* polarizing behavior. [13][14]

This pattern can be explained both by the lack of overall polarization exhibited by these models and the homogeneous distribution of polarization across groups, which is not observable to this extent in human annotations (Fig. 6), and which is further supported by overall small InPol values (App. J; Table 12). Our findings therefore support social science work which warns against generating human annotation data without substantial replication studies (Rossi et al., 2024; Kapania et al., 2025). See App. J for the ablation experimental procedure, full results, model details, dataset selection, exploratory analysis, and comparison to human annotations.

Finding 5. *LLMs with persona-prompting do not produce annotations with attributable polarization across any group, which may be explained by the even distribution of polarization across groups.*

7 Conclusion

We discovered issues regarding the presence of inherent and conflicting polarization which render current methods around subgroup disagreement attribution suboptimal in realistic circumstances. We created a new metric that overcomes these issues and outperforms alternatives methodologies in realistic annotation settings, while also quantifying the attribution of each group. By applying this

[13]These prompts do however tend to increase InPol, although not to human levels—see App. J; Table 12.

[14]We note that there may exist other ways to induce polarizing behavior in LLMs such as Supervised Fine-tuning, although this is outside the scope of this work.

metric to four subjective NLP datasets, we discover that (1) ethnicity and gender seem to generally drive polarization attribution, and (2) ideological groups may be more important than reported in the literature. Finally, we find that persona prompting is not capable of unlocking consistent subgroup disagreement even when using adversarial instruction prompts.

AI USE STATEMENT

In this work we used proprietary models such as Claude's Sonnet 5 and OpenAI's GPT-5 for partially implementing the synthetic experiments in §5.2 and generating parts of the library, experiment and graph-creation code. All such tasks were executed with direct author supervision, and all relevant code files (library, experiments, graphs) have been edited manually at multiple points by the authors to ensure reproducibility, accuracy and software quality. Additionally, we attempted to use Claude's Sonnet 5 for proposing research directions for this work, but ended up discarding all suggestions. We also used Scholar Labs to find citations for targeted claims (e.g., for known issues with persona prompting), although all such citations were manually filtered by us. Finally, we used a locally hosted Gemma-4 model (specifically the `gemma-4-E4B-it-Q4_K_M.gguf` variant) for rephrasing this manuscript in order to improve readability. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

We ensure that our LLM annotation study is reproducible by selecting exclusively open-source, locally hosted models and fixing all relevant seeds. An ablation study where we repeat annotations multiple times shows that our setup achieves 100% consistency across all models. In order to ensure that the LLM annotators are not influenced by minute differences in the prompts, we execute a second ablation study where we rephrase the main annotation prompt three times and compare outputs. This procedure revealed that two models were especially sensitive to such changes, which led us to exclude those models from the analysis presented in this manuscript. Information on the synthetic experiments used in §4,5 is disclosed in App. G. We analyze software design decisions, limitations and computational details in App. D and elaborate on our experimental setup in App. E, while ablations around sample sizes and sample selection for the Kumar dataset are detailed in App. H. Additionally, we include a discussion around the use of current NLP datasets, as well as an empirical estimation of the number of required annotators for reproducible results in App. I. All code, data-processing scripts and the library implementing *apunim* are released anonymously: library `https://anonymous.4open.science/r/apunim-5F70`, online (HTML) documentation `https://anonymous.4open.science/r/apunim-page`, replication code `https://anonymous.4open.science/r/aposteriori-unimodality-E8C1`.

ETHICS STATEMENT

Dual Use Concerns The primary objective of this study is to identify systemic sources of polarization in subjective NLP datasets to facilitate the construction of more robust and equitable AI systems. However, given that our methodology allows for the quantitative attribution of specific judgment patterns to sociodemographic and ideological characteristics, we acknowledge three primary risks inherent in this research.

First, *Discriminatory Profiling*: If the metric is applied outside of a dataset quality analysis, its ability to correlate specific judgment patterns with annotator groups could be misused for surveillance or unjust profiling. Second, *Automated Bias Amplification*: The diagnostic patterns of polarization we detect are reflections of human systemic biases. Without human oversight, training subsequent models on these patterns risks automating and entrenching existing societal prejudices. Finally, the knowledge provided by *apunim* could be *weaponized*, allowing malicious actors to strategically target weak points in moderation systems or intentionally maximize societal conflict.

We stress that our metric functions strictly as a diagnostic warning signal. Any deployment in a real-world system must mandate a rigorous human-in-the-loop review, ensuring that identified patterns lead to systemic improvements (e.g., policy reform, demographic diversity initiatives) rather than merely justifying an automated system.

Data Sourcing and Privacy All datasets employed in this study were either publicly accessible or provided to us by the original authors with full disclosure regarding the purpose of our research. No identifiable, personal information was included in our pipeline. We do not release proprietary data provided by the authors, nor have we provided any to proprietary LLMs. Finally, we note that our data is limited to English posts from a U.S. cultural context and our results may not be representative for other cultures.

Examples Used Our study utilizes examples of comments that may be considered divisive across certain demographics. These examples were generated by our team solely for demonstration purposes. We explicitly state that the contents of these examples and the groups studied do not necessarily reflect the views or behaviors of the groups they represent, nor do they reflect the opinions of the authors.

Prompts Used This study uses adversarial prompts which explicitly instruct the model to change its outputs according to their persona prompts. The purpose of these prompts is to test the LLMs under an extreme case of prompt engineering and are not to be used for generalized usage. The use of these prompts may degrade model performance, increase the risk of stereotyping and produce misleading results.

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A Acronyms

InPol Inherent Polarization
nDFU normalized Distance From Unimodality
NLP Natural Language Processing
GPLv3 GNU General Public Licence v3
AA African American
PC Personal Characteristic
HS Hate Speech
SD Standard Deviation

B Exploratory Analysis

The number of annotators per item for all datasets presented in §3 can be found in 8, and descriptive statistics can be found in Table 3. The DICES-990 dataset has exactly 123 annotations per item, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few items that vastly exceed the promised 5 number of annotators. We do not include these items, since they are very likely a data error. Tables 4,5,6,7,8 show the annotation counts for each PC group in each dataset.

C Statistical Estimation

Since annotation histograms may be noisy, we additionally assess whether the observed apunim value could arise by chance. To assess statistical significance, we construct a null distribution of apunim values from the randomized partitions and compare the observed apunim against this distribution using a one-sample Student-t test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

$$\begin{aligned} H_0 &: \bar{apunim}_{\mathrm{rand}}(d, g) = apunim(d, g) \\ H_a &: \bar{apunim}_{\mathrm{rand}}(d, g) \neq apunim(d, g) \end{aligned} \tag{8}$$

where $\bar{apunim}_{\mathrm{rand}}(d, g)$ denotes the average apunim values obtained from t randomized annotator partitions.

We account for the presence of multiple subgroups in each PC dimension using a multiple-hypothesis correction. Since our test is parametric (§C), it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per PC group to estimate nDFUs in Equations 4-6, (3) the apunims of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §3). We discuss condition (2) in App. I. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single group in principle, in practice we often want to test polarization for all subgroups of a specific dimension. Since we are simultaneously considering multiple hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holms method Holm (1979). The test is parameterized by the confidence level of our test, which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses Chen et al. (2017)).

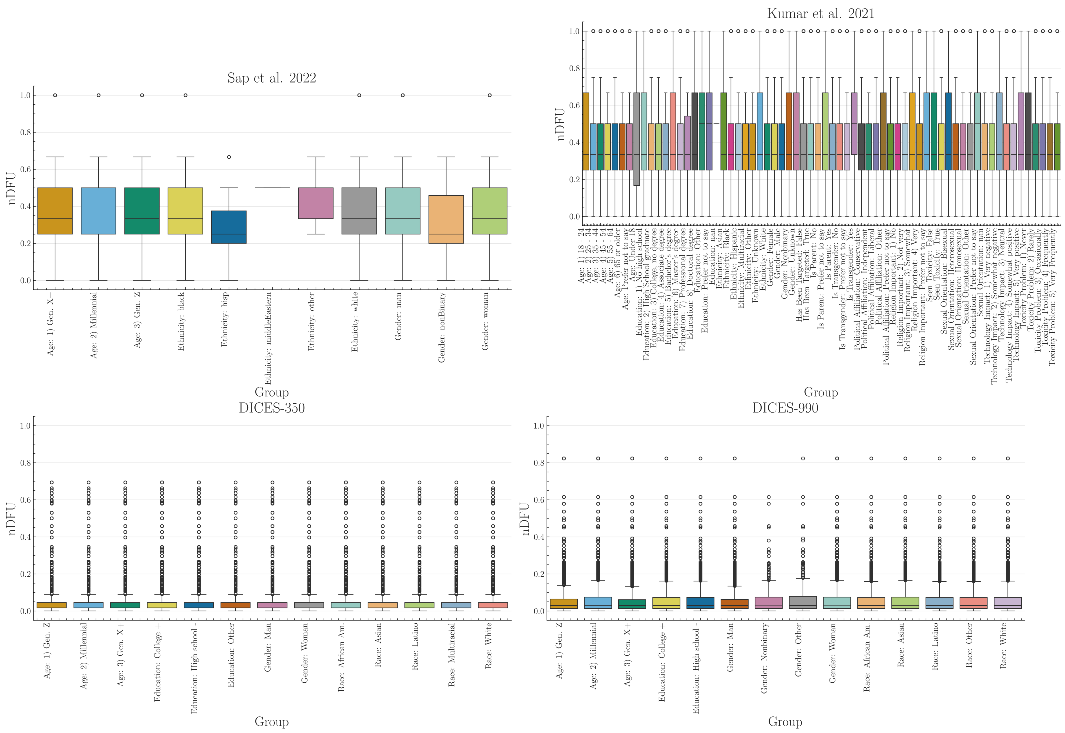

Figure 7: Boxplots of dataset polarization for the human datasets considered in this paper for all groups in each examined PC dimension.

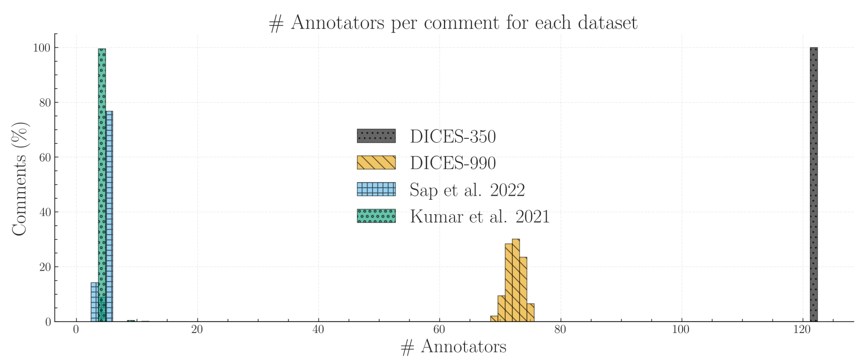

Figure 8: We calculate the ratio of items for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few items in the Kumar dataset exceeding this threshold (see Table 3), which is likely caused by a data error.

D Software

D.1 Installation and Usage

We release the `apunim` package, which is a Python library that implements both the nDFU and apunim calculations, as well as its statistical test. The software is available on PyPi We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (`https://anonymous.4open.science/r/apunim-page`).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project's repository (`https://anonymous.4open.science/r/`

Dataset	count	mean	SD	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8	1.17	69	72	73	74	76
Sap	585	5.6	.78	3	6	6	6	12
Kumar	2998	5.02	.34	5	5	5	5	10

Table 3: Number of annotations per item in each of the four datasets used in this study.

Category	**Subgroup**	**Support**
Gender	Woman	21700
	Man	21350
Race	White	10500
	African Am.	10150
	Asian	9100
	Latino	7700
	Multiracial	5600
Age	1) Gen. Z	19600
	2) Millennial	12600
	3) Gen. X+	10850
Education	College +	26250
	High school -	14350
	Other	2450

Table 4: Subgroup Counts for DICES-350. “Support” refers to annotation count.

`aposteriori-unimodality-E8C1`). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy, scipy, statsmodels`).

D.2 Implementation Details

By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple PC dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{r}_i$, which are originally calculated to obtain the apriori polarization values (see Eq. 4), for the p-value calculations (§C). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{r}_i$ in Eq. 4), and the number of apunim values used in the p-value calculation have to be the same.

We use version `1.0.6` of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Eq. 4) and the p-value estimation (§C), and set a confidence level of $\alpha = 0.05$.

While the library efficiently handles large numbers of annotators through vectorized operations, performance degrades significantly when processing a high volume of items, particularly when annotators belong to multiple distinct groups. For instance, while experiments across all datasets in §3 complete within a minute, the Kumar dataset necessitates approximately one hour of computation using commercial hardware. Furthermore, the annotator variance analysis detailed in App. I required over ten hours. While this latter time requirement is anticipated due to the inherently demanding nature and non-standard application of our library, it demonstrates the existence of a potential scaling bottleneck. The same applies to a lesser extent for the subsampling experiments on the DICES datasets, taking collectively three hours of computation.

Category	Subgroup	Support
Gender	Woman	36050
	Man	35392
	Nonbinary	331
	Other	330
Race	Other	26394
	Asian	19885
	White	12322
	Latino	7170
	African Am.	6332
Age	Gen. Z	14899
	Millennial	41542
	Gen. X+	15662
Education	College +	64054
	High school -	8049

Table 5: Subgroup Counts for DICES-990. "Support" refers to annotation count.

Category	Subgroup	Support
Age	Gen. Z	252
	Millennial	2259
	Gen. X+	786
Ethnicity	white	2168
	black	1091
	hisp	32
	other	5
	middleEastern	1
Gender	man	1859
	woman	1412
	nonBinary	26

Table 6: Subgroup Counts for Sap et al. 2022. "Support" refers to annotation count.

We hypothesize that this bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each item, for each PC dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each PC subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` package Tange (2025) for our own experiments). We also note that the Kumar dataset also requires substantial RAM resources, estimated around 42–46GB of dedicated RAM should the whole dataset be considered (this is not a requirement for loading the sample used in this study). The CPU-bound experiments require approximately 3 hours to run, assuming a computer capable of efficient multiprocessing.

Finally, we note that in Fig. 11, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of items in DICES-990 were annotated by more than 73 annotators—see App. B).

E Design Decisions

E.1 Defining Apriori Polarization

In §4.1, we explained how our methodology needs to compare the polarization of $A(c)$ and $A(c \mid g)$. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset—in this case, standard practice dictates comparing the subset with its complement; i.e., the

Category	Subgroup	Support
Gender	Female	8011
	Male	6832
	Unknown	124
	Nonbinary	93
Ethnicity	White	10699
	Black	1940
	Asian	862
	Multiracial	664
	Hispanic	418
	Other	247
	Unknown	230
Age	Under 18	3
	18 - 24	1711
	25 - 34	5969
	35 - 44	3785
	45 - 54	1977
	55 - 64	1101
	65 or older	471
	N/A	43
Education	No high school	71
	High School graduate	1353
	College, no degree	3090
	Associate degree	1676
	Bachelor’s degree	6071
	Master’s degree	2261
	Professional degree	236
	Doctoral degree	198
	N/A	71
	Other	32
Sexual Orient.	Heterosexual	12391
	Bisexual	1665
	Homosexual	515
	N/A	302
	Other	131

Table 7: Subgroup Counts for the Kumar et al. 2021 dataset. Part 1 of 2. “Support” refers to annotation count.

in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Fig. 2, left, we would end up subtracting the nDFUs of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

E.2 Filtering Ineligible items

Because *apunim* is fundamentally designed to quantify polarization, we exclude items that exhibit no measurable polarization ($nDFU(c) = 0$). Furthermore, we also exclude items annotated exclusively by a single group (i.e., items where there are no two groups with at least three annotators each), since we require valid histograms for the non-parametric estimation of nDFU (see Eq. 1). This is an unfortunately common scenario, since many NLP datasets feature only $5 - 6$ annotations per item. In a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split.

Category	Subgroup	Support
Is Transgender	No	14549
	Yes	363
	N/A	148
Politics	Liberal	5997
	Independent	4143
	Conservative	3981
	N/A	609
	Other	330
Is Parent	Yes	7889
	No	7015
	N/A	156
Tech Impact	Very negative	149
	Somewhat negative	1411
	Neutral	1905
	Somewhat positive	7690
	Very positive	3905
Tox Problem	Never	869
	Rarely	2485
	Occasionally	5307
	Frequently	4441
	Very Frequently	1958
Rel Important	No	4735
	Not very	1808
	Somewhat	3548
	Very	4741
	N/A	228
Seen Toxicity	True	11505
	False	3555
Has Been Target	False	10741
	True	4319

Table 8: Subgroup Counts for Kumar et al. 2021 dataset. Part 2 of 2. "Support" refers to annotation count.

F WHAT IS INPOL?

Why Does it Exist? Unlike disagreement, InPol does not provide any information, but rather represents noise or uncertainty translated in terms of polarization. In order to minimize it, we need to find its cause. Since the DICES datasets are the only ones without *any* items with InPol leads us to two hypotheses:

Hypothesis 1 (Low annotator count)**.** A low number of annotators per item may provide our model with insufficient information.

Hypothesis 2 (Inaccessible groups)**.** InPol may arise from the presence of partially represented groups within the annotator pool which may go undetected if they are represented by only one or two annotators, as nDFU is undefined for these groups.

Both of these explanations are plausible and reflect known limitations of most NLP datasets. Unfortunately, we can not confirm any of these hypotheses

The fact that the DICES datasets (featuring 70+ annotators per item) exhibit zero InPol in all items may initially seem to validate either of the above hypotheses. However, even when repeating the experiment using random annotator subsets (ranging from 5 to 50 annotators) and taking the maximum across ten random samples for each subset size, *we still observed zero InPol* (App. I). This

Comment	InPol
The minks mentioned in the headline are indeed "rising" from where they were based, but that is due to them being buried in shallow mass graves (Africa) and filling with gas that causes them to rise out. Rest assured, they're still very much dead, thankfully!.	1
There should be such thing as a bitch resistance for us women	1
There's a big difference between being a bitch and being a woman	1
i just watched a video of @joisnotonfire being an annoying hoe on skype i'm laughing	1
niggas love to commit crime	1
I swear to God above I love ya and when ya be with this brotha, I swear I love you my bro as long as you not a nigga	0.5
I will become a cold ass bitch because of my treatment	0.5
I would happily lay out a nigga	0.5
Shoot yourself in the head you hypersensitive fuckin bitch.	0.5
not all women are bitches, but all bitches are women	0.5
" you ugly bitch!!!"- that seems to be my real name these days"	0
"I am happy to report that drowning does kill this demonic entity. Oh holy are the waters of Chernarus!"" Omg, DED ;)"	0
"Let them praise the name of the LORD: for his name alone is excellent; his glory is above the earth and heaven."" Psalms 148:13 #Bible"	0
someone plz get this nigger some help. He's playing up like mad	0
stop talking you fucking nigger !	0

Table 9: Sample of comments from the Sap dataset and their InPol. Note that unambiguously toxic comments can be assigned with zero InPol, while more nuanced cases tend to have higher values.

finding indicates that the DICES datasets may be structurally different. This explanation is plausible given that the annotation targets are LLM texts, which may be relatively unsubtle compared to human items, as supported by low overall polarization (Fig. 7; App B). Although a definitive answer to this question eludes us due to the lack of other datasets with high annotator counts per item, the indication that the DICES datasets may not feature similar polarization patterns is noteworthy.

How Does it Look Like? Table 9 shows a random sample of items taken from the Sap dataset and their respective InPol values (§4.2). Note that explicitly toxic or racist items tend to have InPol values of zero, alongside clearly unproblematic items. Inversely, more ambiguous and less comprehensible items tend to have larger InPol values. This pattern may suggest that InPol may be tied to the overall ambiguity of the statements, especially in cases where items may either be interpreted as very toxic, or non-toxic, which agrees with the overall goal of nDFU (§2.2). We include the InPol values for each individual item for all four datasets in the supplementary material.

G Synthetic Experiments

This appendix details the synthetic experiments used to illustrate the functionality and behavior of nDFU. In all simulations, the bin size is set to $N_{bins} = 10$.

Disagreement vs. Polarization To explore the relationship between annotation polarization and disagreement in Fig. 1, we constructed a 2×2 matrix of synthetic datasets, each drawing from a pool of $N_{annotators} = 100$. Polarization is simulated by defining a minority group, which constitutes 20% of the total annotators (n_special $= 20$). These minority annotators are assigned a targeted mean shift (mean_shift $= 1$) from the base distribution. The base variance for these distributions is set at variance $= 0.3$. Different types of distributions (unimodal, bimodal, multimodal, uniform) are achieved by controlling the base_means and applying variance multipliers (e.g., variance $\times\, 3$ for multimodal).

Single-Item Polarization We simulate the distribution of toxicity annotations in Fig. 2, left, across two demographic groups (Men and Women). The distribution of Men's scores is modeled using a

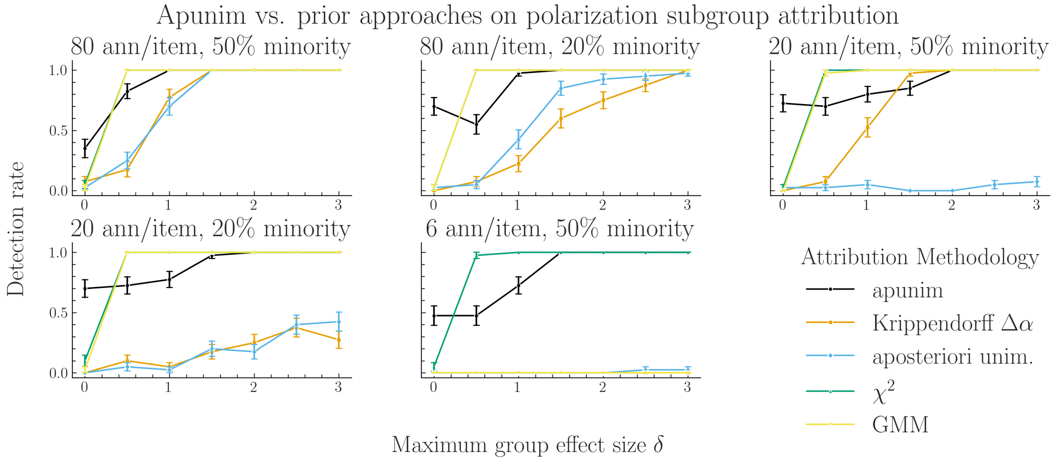

Figure 9: Polarization group attribution on randomly generated, synthetic distributions. Bars indicate standard error. For each experiment we assign a number of annotators per item, and split the annotator groups 50-50 or 80-20. The setup is much more simplified, relying on fixed-mean differences instead of the setup used in §5.2.

truncated normal distribution with loc $= 2$ and scale $= 1.3$, while Women's scores are modeled with loc $= 8$ and scale $= 1.3$. The combined dataset, representing all annotations, is used to demonstrate the impact of population weighting on the overall nDFU score. The sample size for each group in this visualization is $N_{annotators} = 50$.

Discussion-Level Polarization We examine nDFU scores for three specific textual examples presented in Fig. 3. For the individual items, $N_{annotators} = 200$ for each group. In the first case (misogynist item), Men's scores are modeled with loc $= 6$, while Women's scores are modeled with loc $= 2$. In the second case (misandrist item), the roles are reversed. The third visualization combines distributions for men and women derived from the respective single-item distributions.

Synthetic Polarization Datasets To model polarization more realistically for the experiments in §5.2, we use a item-specific group effect rather than assuming that annotators in one group are uniformly biased toward higher or lower ratings. Specifically, annotations are generated according to $y_{ij} = b_j + g_i p_j + \epsilon_{ij}$, where b_j represents the baseline rating for item j, $g_i \in \{-\frac{1}{2}, +\frac{1}{2}\}$ indicates the annotator's group, and p_j controls the magnitude and direction of polarization for item j. We sample p_j uniformly from $[-\delta, \delta]$, such that different items can favor different groups while δ controls the maximum group-level separation. Thus, rather than modeling a fixed group bias, the simulation captures situations where groups systematically disagree on particular items, with the direction and strength of disagreement varying across items but being consistent among annotators of the same group.

To determine whether an observed score is statistically significant, we construct a null distribution for each method. The null hypothesis is that the group labels are unrelated to the observed annotations. We therefore repeatedly shuffle the group labels across annotators, recompute the method's statistic, and use the resulting scores to form the null distribution. The observed score is then compared with this distribution. See App. C for the apunim p-value implementation.

H Experiment Ablations

H.1 Why Do Statistical Methods Fail?

In §5.2 we mentioned that the under-performance of the χ^2 and GMM methodologies is likely caused by opposing polarization effects. We compare this simulation model with a simplified simulation that imposes a fixed polarization effect across all items, modeling the scenario where group bias is uniform. Specifically, the polarization shift$_{ij}$ is constant across j: shift$_{ij} = g_i \cdot \delta/2$. In this simplified case, the group difference is a global shift, meaning that if Group B rates one item higher than Group A, they rate all items higher.

The experiment, whose results are shown in Fig. 9 shows that these methods do indeed show excellent performance. This is expected, since they rely on consistent fixed means effects, which is disrupted in our original formulation of the problem. Thus, while these methods perform admirably in ideal conditions, they may completely fail at realistic annotation datasets.

H.2 How Does Dataset Size Influence Apunim?

Table 10: Comparison of apunim values across different sample sizes (1k, 10k, 30k) for the Kumar dataset.

PC	**1000**	**10000**	**30000**
Age			
8 - 24	-0.002	-0.037	-0.016
25 - 34	0.045	0.000	0.000
35 - 44	-0.043	-0.001	-0.001
45 - 54	0.001	0.013	0.003
55 - 64	0.002	-0.057	0.001
65 or older	-0.007	N/A	0.948
Education			
High School graduate	-0.023	-0.005	0.014
College, no degree	-0.045	0.010	0.006
Associate degree	0.001	0.005	0.009
Bachelor's degree	0.029	-0.001	-0.000
Master's degree	0.024	0.004	-0.000
Ethnicity			
Asian	-0.012	0.011	0.007
Black	0.036	-0.004	-0.012
Hispanic	0.004	0.000	-0.099
Multiracial	0.008	0.000	0.000
Other	-0.008	N/A	-0.306
White	-0.023	0.000	0.000
Gender			
Female	-0.006	0.002	0.001
Male	0.040	-0.002	-0.001
Has Been Targeted			
False	-0.059	0.000	0.000
True	0.017	-0.003	-0.002
Is Parent			
No	-0.084	0.001	0.001
N/A	0.004	1.000	0.000
Yes	0.071	0.000	0.000
Is Transgender			
No	-0.102	0.000	0.000
Political Affiliation			
Conservative	0.056	-0.003	-0.002
Independent	-0.003	0.001	0.002
Liberal	-0.026	0.001	0.001

Other	-0.003	0.000	0.000
Religion Important			
No	-0.071	0.002	0.001
Not very	-0.013	0.010	0.006
Somewhat	0.022	0.003	-0.002
Very	0.043	-0.003	-0.002
Seen Toxicity			
False	0.025	-0.001	-0.001
True	-0.082	0.001	0.000
Sexual Orientation			
Bisexual	-0.003	-0.040	-0.013
Heterosexual	-0.067	0.000	0.000
Homosexual	-0.012	0.000	0.053
Technology Impact			
Somewhat negative	-0.008	0.000	0.003
Neutral	0.017	-0.014	0.000
Somewhat positive	-0.053	0.000	0.001
Very positive	0.027	-0.004	-0.003
Toxicity Problem			
Never	-0.008	-0.037	-0.081
Rarely	0.004	0.003	-0.001
Occasionally	-0.012	-0.003	-0.001
Frequently	-0.032	-0.001	0.000
Very Frequently	-0.017	0.030	0.015

Since our approach was designed for the small-scale datasets common in NLP, we were cautious when testing it on the massive Kumar dataset. Because of the data volume, we analyzed various smaller samples (1,000, 10,000, and 30,000 items), shown in Table 10 (rows with more than one N/A values removed). As sample size increased, the statistical metrics (apunim values) consistently shrank, and we found no statistically significant effects in the larger samples.

The test is overly conservative, meaning it almost always suggests there is no significant effect, even when an effect likely exists in reality. This inability to detect a real signal is a Type II Error (a false negative). This suggests that our test is statistically weak or "underpowered" in massive settings. Since our methodology is built for the small-scale NLP datasets that dominate the research field, we deem this limitation acceptable for the scope of our work.

H.3 How Does Sample Selection Influence Apunim?

Astute readers may notice that some variability does exist across differently sized samples for the same examined groups, where for example the sign of the apunim value may switch. We note that the statistically significant values (bold rows) change signs only when the apunim value approaches zero in the largest samples. Thus, the results are not inverted, but rather pushed towards zero.

In order to verify that this variability is caused by the sample size, and not by different samples being drawn, we execute a further ablation study. Specifically, we execute the apunim study on the 3000-sized sample of the dataset, using different subsets of the Kumar dataset. The results can be found in Fig. 10, where we note that variability is not an issue, except for a couple of cases. These cases seem to represent sparse annotator groups, where missing a few items may genuinely change the value of the apunim metric. These groups however are largely not detected by apunim as significant either way (see §6).

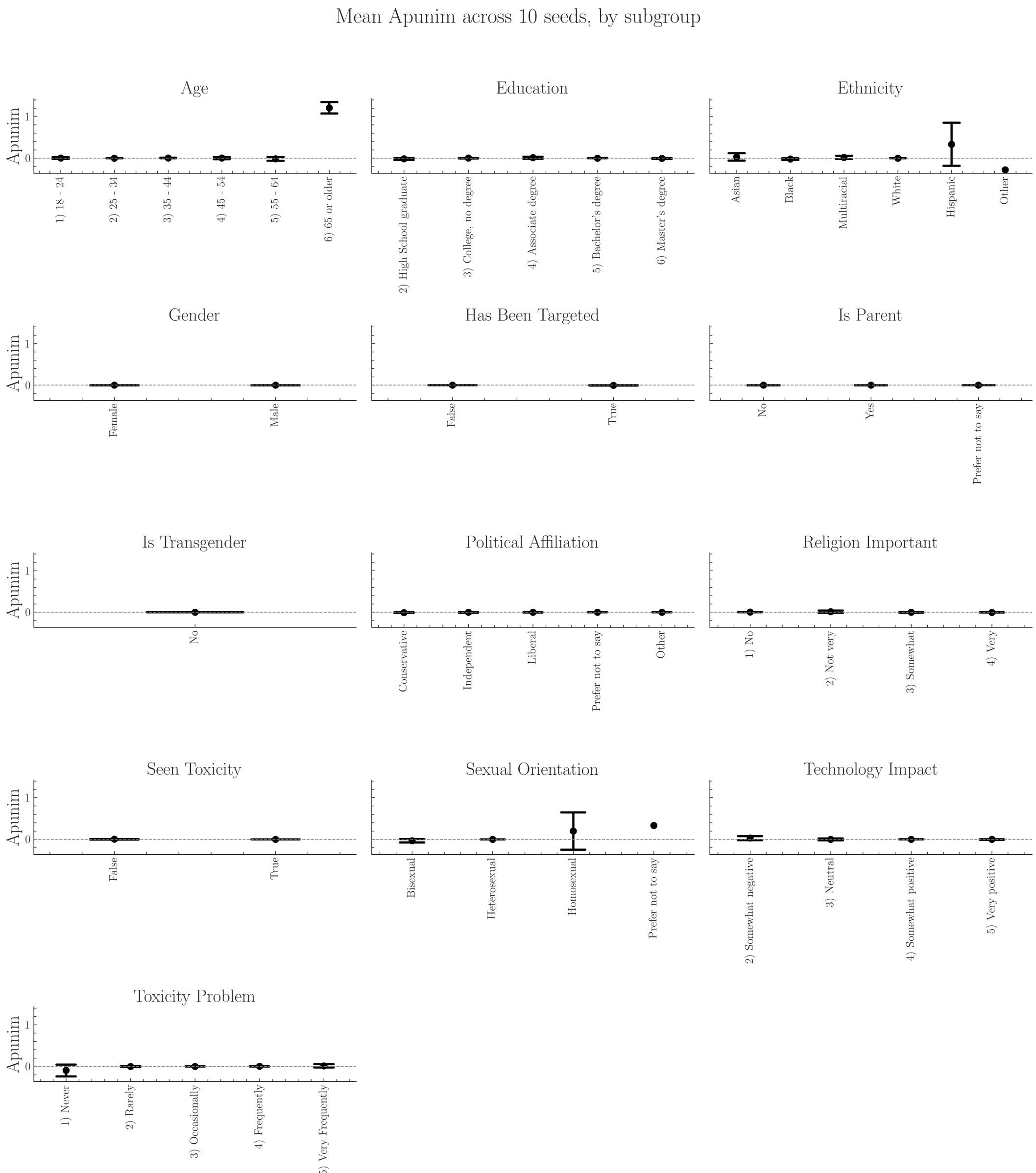

Figure 10: Apunim values on the Kumar dataset ($n_{samples} = 3000$) across ten runs with different subsets. Black bars indicate two SDs. PC dimensions with many groups are more volatile, although all dimensions broadly keep the same apunim sign across subsets.

I The Ideal Subjective Dataset

We briefly discuss the type of data most suited for our proposed methodology, and connect it with current NLP datasets.

How Many Annotators? Our methodology does not directly require numerous annotators overall, only a sufficient number per item—indeed, statistically significant results can be obtained with as little as four annotations (see Sap-AA; Table 19). A large annotator pool however is required to get sufficient coverage on the examined groups. More annotators allow both higher fidelity of each PC dimension, and more such dimensions.

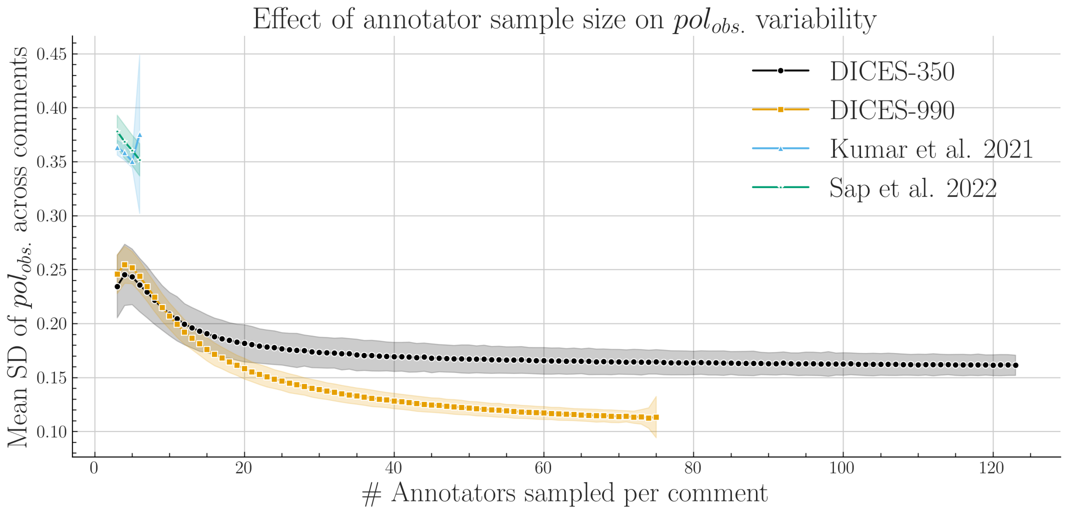

Figure 11: SD of pol_{obs} across items, averaged across 1000 runs (Eq. 6). Shaded areas represent two SDs. Pol. estimates benefit from more annotators up to $n = 20$, at which point they begin to plateau.

How Many Annotators per item? We investigate observed polarization against varying numbers of annotators. Specifically, for every item, we repeatedly sampled n annotators with replacement, calculated pol_{obs} for each sample and measured SD, averaging it across 1000 same-sized samples. As shown in Fig. 11, pol_{obs} exhibits a clear, inverse relationship with the number of annotators, attributed to better histogram estimation provided by nDFU (§5), though the rate of gain substantially diminishes after 20 annotators. Interestingly, some variability persists even with a very high annotator count.

How Many Items? Our methodology is designed to work with smaller datasets common in NLP. Three of the four datasets in the study feature low (300–600) item counts, while the Kumar dataset has been subsampled to 3% of its original size. Large datasets should be used only to provide larger coverage on the target domain, since a much better representation of individual groups can be achieved with more annotators per item, rather than more items. Smaller datasets may also be more practical for social science analysis Rossi et al. (2024).

Filtering Datasets with Desirable Levels of Subjectivity In order to investigate whether the numerous annotators per items can explain the lack of InPol in the DICES datasets (§4) we gradually subsample the annotators for each item randomly with replacement, starting from five annotators per item, and scaling up to the full complement of annotators in the dataset with a step of two. For each such step, we choose ten annotator samples and compute the maximum InPol across all runs. Even with annotator counts similar to the Kumar and DICES datasets, InPol remains zero, suggesting that the absence of InPol can not be attributed to annotator number alone—rather, it is plausibly a pattern attributable to the DICES datasets themselves. This kind of analysis may be used in the exploratory phase of dataset selection in order to filter out datasets without enough group-specific subjectivity.

J LLM Annotation

J.1 Models Used

We use only open-source models, as proprietary models such as OpenAI's GPT-5 yield much less reproducible results (Bisbee et al., 2024). Furthermore, the use of open-source models requires much fewer resources, enabling research on this task with a less demanding budget, as well as scalable deployment in real-world settings (Rossi et al., 2024). It is also worth noting that since the datasets used are not publicly licensed, we can not risk leaking the data contained within to proprietary models. Generally, the use of local models protects the personal data of the users contained in our datasets, ensures that our research can be replicated in domains where anonymity must be maintained, and renders our work compatible with existing legislation European Union (2024; 2016). Table 11 shows details on the LLMs used for this study. We specifically use 4-bit quantized versions from the `unsloth huggingface` repository.

Family	Version	#Params
`OLMo` (Groeneveld et al., 2024)	`2 (0325)`	32B
`Qwen` (Yang et al., 2025)	`2.5`	32B
`LLaMa` (Grattafiori et al., 2024)	`3.3`	70B
`OLMo` (Olmo, 2026)	`3`	7B
`Qwen` (Yang et al., 2025)	`2.5`	7B
`LLaMa` (Grattafiori et al., 2024)	`3.1`	8B

Table 11: LLMs used in this study.

J.2 Experimental Setup

Since each of the datasets in §3 was annotated using different annotation guidelines and targets we use different instruction prompts for each dataset. The "default" annotation prompt is directly extracted from the studies in the case of Sap and Kumar, and inferred from the instruction guidelines used in the DICES datasets with minor modifications. We use standard persona prompting procedures sampled randomly from the available PC dimensions in each respective dataset. Six LLM annotators are selected for each item and model.

The prompts used in the ablation study were provided by a GPT-5 and a local Gemma-4 (quantized) model by providing the original prompt and instructing it to produce four variants. The tested prompts use different methods to restrict the model's behavior. The initial prompts ("*Persona*"require the model to adopt a persona's viewpoint to judge the comment. Other prompts ("*direct*") treat the persona's characteristics as an absolute, non-negotiable rule that overrides standard moderation guidelines. Another approach ("single") compels the model to perform a persona-based check before submitting its final rating. The most aggressive prompts ("stereotype") force the model into extreme decision spaces. The full main and ablation annotation prompts can be found in App. L.

We use a single Quadro RTX 6000 GPU for tasks requiring small (7-8B) models and two such GPUs for the rest. The annotation task, including ablations, requires approximately 300 hours of computation to complete. We use the `transformers` library to load and manage the LLMs. We release the relevant code in the supplementary material. Our codebase is platform-independent except for three high-level bash scripts and a full environment file is included for reproduction.

J.3 Exploratory Analysis

Table 12: Mean inherent polarization (mean ± 2 SD) per (dataset, prompt), for Human and each LLM.

Dataset	Prompt Source	default	stereotype	persona
kumar	Human	0.167 ± 0.472	0.168 ± 0.473	0.168 ± 0.473
	llama70b	0.001 ± 0.032	0.000 ± 0.000	0.001 ± 0.045
	olmo32b	0.009 ± 0.137	0.029 ± 0.236	0.021 ± 0.203
	qwen32b	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
	qwen7b	0.005 ± 0.095	0.009 ± 0.137	0.003 ± 0.083
sap	Human	0.097 ± 0.420	0.097 ± 0.420	0.097 ± 0.420
	llama70b	0.000 ± 0.000	0.002 ± 0.058	0.002 ± 0.058
	olmo32b	0.007 ± 0.115	0.015 ± 0.171	0.013 ± 0.161
	qwen32b	0.000 ± 0.000	0.002 ± 0.058	0.000 ± 0.000
	qwen7b	0.000 ± 0.000	0.002 ± 0.058	0.000 ± 0.000

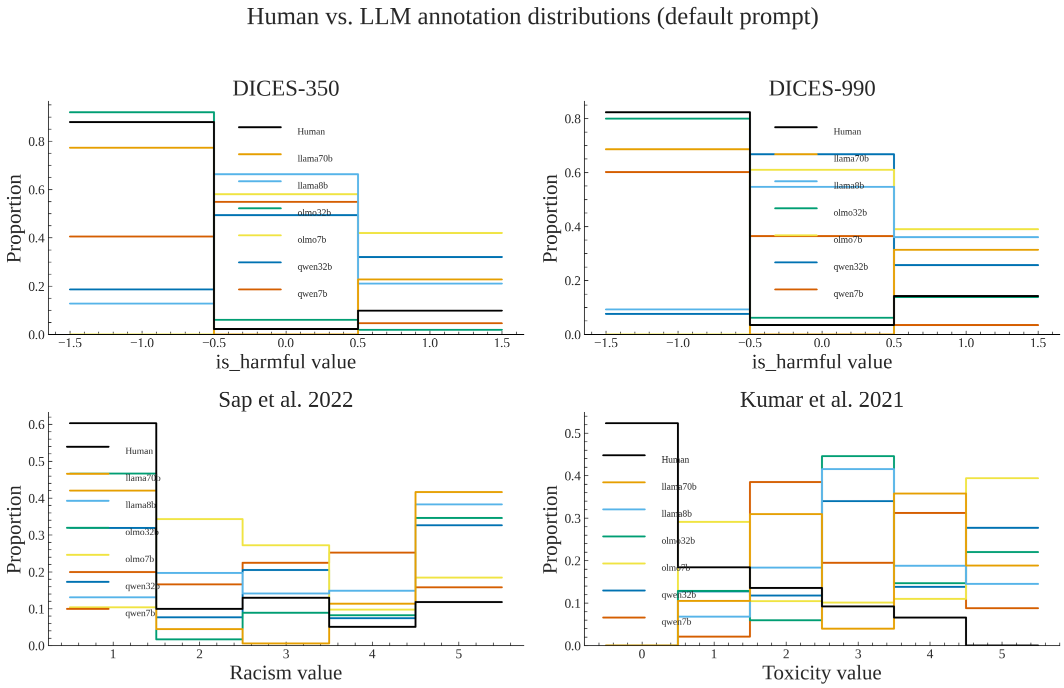

Figure 12: Annotation frequencies for humans and each LLM in the primary annotation task using the default prompt.

Fig. 12 shows the different annotation frequencies of LLMs and human annotators in the same dataset split. Across all four datasets, we observe a consistent mismatch between human and LLM annotation distributions. For the DICES-350 and DICES-990 harmfulness tasks, human annotations are heavily concentrated at the lowest harmfulness value, whereas most LLMs assign substantially more probability to moderate and high harmfulness values. A similar pattern appears in the Sap et al. (2022) racism dataset: human judgments are concentrated at low racism scores, while LLMs produce broader distributions with considerably more mass at higher scores, particularly values 4–5. The (Kumar et al., 2021) toxicity results show the same general trend, with humans assigning most probability to zero toxicity while LLMs distribute their predictions across higher toxicity levels, often peaking around values 2–4. Importantly, the magnitude of this discrepancy varies across models; some LLMs are closer to the human distribution, while others show substantially stronger shifts toward higher scores.

We also investigate whether LLM annotations include any InPol. Table 12 shows that InPol is significantly smaller than in human annotations. Adversarial prompts seem to consistently increase it, although it does not approach human levels. It is also notable that as with LLM apunim values (next section), InPol seems to be primarily model-dependent.

J.4 Annotation Ablations

We examine the following conditions for selecting models and datasets for our LLM annotation experiment. All ablation experiments use a sampled 20% fraction of the full LLM annotation dataset used in the main text.

The LLMs should be consistent given the same prompt While we have fixed all relevant seeds, it is possible that generation deviates across repeated model prompting given the same input (Bisbee et al., 2024). We check this condition by providing the model with the main annotation prompt (see App. L) five times and find that generation is indeed deterministic under our setup for all models (Table 13).

Table 13: Consistency (Krippendorff's α, ordinal) of each model with itself across repeated runs of the same (default) prompt.

Dataset	Model	Runs	Items	Krippendorff's alpha
DICES-350	llama70b	5	174	1.0000
DICES-350	llama8b	5	180	1.0000
DICES-350	olmo32b	5	180	1.0000
DICES-350	olmo7b	5	173	1.0000
DICES-350	qwen32b	5	167	1.0000
DICES-350	qwen7b	5	180	1.0000
DICES-990	llama70b	5	180	1.0000
DICES-990	llama8b	5	180	1.0000
DICES-990	olmo32b	5	180	1.0000
DICES-990	olmo7b	5	180	1.0000
DICES-990	qwen32b	5	180	1.0000
DICES-990	qwen7b	5	180	1.0000
Sap et al. 2022	llama70b	5	180	1.0000
Sap et al. 2022	llama8b	5	180	1.0000
Sap et al. 2022	olmo32b	5	178	1.0000
Sap et al. 2022	olmo7b	5	172	1.0000
Sap et al. 2022	qwen32b	5	180	1.0000
Sap et al. 2022	qwen7b	5	180	1.0000
Kumar et al. 2021	llama70b	5	600	1.0000
Kumar et al. 2021	llama8b	5	600	1.0000
Kumar et al. 2021	olmo32b	5	600	1.0000
Kumar et al. 2021	olmo7b	5	567	1.0000
Kumar et al. 2021	qwen32b	5	600	1.0000
Kumar et al. 2021	qwen7b	5	589	1.0000

The LLMs should be prompt insensitive Small changes in the prompt have been shown to drastically affect LLM output in some tasks depending on the experimental setup Razavi et al. (2025). In order to verify whether any models are unreliable under our setup, we create two alternative formulations of each prompt without a change in substance using a GPT-5 model (free version), whose outputs we manually verified (see App. L). Table 14 shows that while most LLMs are relatively robust to changes in formulation, the smaller ones (OLMo-7B and LLaMa-8B) are not. They are thus excluded from the annotation study.

The annotation guidelines should be unambiguous The studies that provided the datasets used in our work (see §3) do not include LLM annotation, nor equivalent LLM prompts, which meant we had to generate them from the human annotation guidelines ourselves. This was not a problem in the case of the Sap and Kumar datasets, since the annotation guidelines were short, simple and required minimal edits to be transformed into LLM annotation guidelines. However, the DICES datasets use poorly documented guidelines and the target label we used is an aggregation of multiple annotation tasks. For these reasons the prompt creation is non-trivial and prone to errors. Indeed, Table 15 shows that inter-model agreement is extremely low for the DICES datasets, even when we remove the sensitive models (Table 16). Given that the DICES datasets are (1) problematic w.r.t. inherent polarization (see App. G), (2) conditioned on several dozen annotations per item (3) difficult to find reliable annotation prompts for LLMs, we decided to exclude them from the annotation study.

J.5 LLM Annotation Results

Tables 17 and 18 show the apunim results for the LLM annotation task on the Kumar and Sap datasets respectively. Apunim values are largely model-dependent, but no other concrete pattern can be discerned. Fig. 13 shows the overall polarization patterns when using the different adversarial instructions. The Qwen-23B and LLaMa-70B models consistently produce no polarization, while the persona-focused prompt seems to elicit some varied polarization, even though it does not translate to higher apunim values. This pattern would indicate that the persona-focused prompt likely

Table 14: Consistency (Krippendorff's α, ordinal) of each model with itself across the three paraphrased prompt variants.

Dataset	Model	Variants	Items	Krippendorff's alpha
DICES-350	llama70b	3	180	0.5531
DICES-350	llama8b	3	180	-0.2432
DICES-350	olmo32b	3	180	0.3822
DICES-350	olmo7b	3	180	0.6254
DICES-350	qwen32b	3	180	0.4212
DICES-350	qwen7b	3	180	0.0091
DICES-990	llama70b	3	180	0.6720
DICES-990	llama8b	3	180	0.0026
DICES-990	olmo32b	3	180	0.6031
DICES-990	olmo7b	3	180	0.7773
DICES-990	qwen32b	3	180	0.5907
DICES-990	qwen7b	3	180	0.2309
Sap et al. 2022	llama70b	3	180	0.8829
Sap et al. 2022	llama8b	3	180	0.6717
Sap et al. 2022	olmo32b	3	180	0.9086
Sap et al. 2022	olmo7b	3	180	0.7090
Sap et al. 2022	qwen32b	3	180	0.9199
Sap et al. 2022	qwen7b	3	180	0.7993
Kumar et al. 2021	llama70b	3	600	0.8406
Kumar et al. 2021	llama8b	3	600	0.3988
Kumar et al. 2021	olmo32b	3	600	0.7827
Kumar et al. 2021	olmo7b	3	593	0.3897
Kumar et al. 2021	qwen32b	3	600	0.9457
Kumar et al. 2021	qwen7b	3	600	0.5578

Table 15: Consistency (Krippendorff's α, ordinal) between LLMs given the same (default) prompt, per dataset.

Dataset	Prompt	Models	Items	Krippendorff's alpha
DICES-350	default	6	1800	-0.0074
DICES-990	default	6	1800	0.0868
Sap et al. 2022	default	6	1800	0.6537
Kumar et al. 2021	default	6	7741	0.4519

adds more overall variance to the persona-conditioned annotations, even though that variance is not consistent.

Table 17: Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across the default, stereotype and persona prompts.

SDB Feature	Value	Model	Default	Stereotype	Persona
Age	1) 18 - 24	llama70b	0.0000	-0.0000	-0.0000
		olmo32b	0.0000	0.0000	-0.0000
		qwen32b	-0.0000	—	0.0000
		qwen7b	0.0000	-0.0000	-0.0000
	2) 25 - 34	llama70b	—	0.0000	0.0000
		olmo32b	-0.0000	0.0000	-0.0000
		qwen32b	0.0000	-0.0000	-0.0000

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Table 17: Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across the default, stereotype and persona prompts.

SDB Feature	Value	Model	Default	Stereotype	Persona
		qwen7b	0.0000	-0.0000	0.0000
	3) 35 - 44	llama70b	-0.0000	0.0000	0.0000
		olmo32b	0.0000	0.0000	-0.0000
		qwen32b	-0.0000	—	0.0000
		qwen7b	0.0000	-0.0000	0.0000
	4) 45 - 54	llama70b	0.0000	0.0000	0.0000
		olmo32b	0.0053	-0.0172	0.0212
		qwen32b	0.0000	0.0000	0.0000
		qwen7b	0.0072	0.0000	0.0000
	5) 55 - 64	llama70b	0.0000	0.0000	—
		olmo32b	0.0000	0.0000	0.0000
		qwen32b	0.0000	0.0000	0.0000
		qwen7b	0.0000	0.0000	0.0000
	6) 65 or older	llama70b	-0.0000	0.0000	0.0000
		olmo32b	0.0291	-0.0022	0.0229
		qwen32b	-0.1662	—	0.0000
		qwen7b	-0.0309	-0.0309	0.0116
	Prefer not to say	llama70b	0.0000	0.0000	0.0000
		olmo32b	0.0157	0.0190	0.0088
		qwen32b	0.2218	0.0000	0.0000
		qwen7b	0.0326	0.0283	0.0151
	Under 18	llama70b	0.0000	—	0.0000
		olmo32b	0.0000	0.0000	0.0000
		qwen32b	-0.0000	—	-0.0000
		qwen7b	-0.0000	-0.0000	0.0000
Education	1) No high school	llama70b	0.0000	-0.0000	0.0000
		olmo32b	-0.0000	0.0000	-0.0000
		qwen32b	0.0000	—	0.0000
		qwen7b	0.0000	0.0000	0.0000
	2) High School graduate	llama70b	0.0000	0.0000	—
		olmo32b	0.0000	0.0000	0.0000
		qwen32b	—	—	0.0000
		qwen7b	0.0000	-0.0000	-0.0000
	3) College, no degree	llama70b	0.0000	—	—
		olmo32b	0.0000	0.0000	0.0000
		qwen7b	0.0000	0.0000	0.0000
	4) Associate degree	llama70b	-0.0000	-0.0000	0.0000
		olmo32b	-0.0000	0.0000	0.0000
		qwen32b	—	-0.0000	-0.0000
		qwen7b	-0.0000	0.0000	-0.0000
	5) Bachelor's degree	llama70b	0.0000	-0.0000	0.0000
		olmo32b	-0.0000	-0.0000	0.0000
		qwen32b	0.0000	0.0000	—
		qwen7b	-0.0000	0.0000	0.0000
	6) Master's degree	llama70b	—	0.0000	0.0000
		olmo32b	—	0.0000	0.0000
		qwen32b	—	—	0.0000
		qwen7b	0.0000	0.0000	0.0000
	7) Professional degree	llama70b	0.0000	0.0000	0.0000
		olmo32b	0.0000	0.0000	-0.0000

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Table 17: Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across the default, stereotype and persona prompts.

SDB Feature	Value	Model	Default	Stereotype	Persona
		qwen32b	0.0000	0.0000	0.0000
		qwen7b	0.0000	0.0000	0.0000
	8) Doctoral degree	llama70b	0.0000	0.0000	—
		olmo32b	0.0000	0.0000	-0.0000
		qwen32b	0.0000	—	0.0000
		qwen7b	0.0000	-0.0000	0.0000
	Other	llama70b	0.0000	0.0000	-0.0000
		olmo32b	-0.0000	-0.0000	0.0000
		qwen32b	—	0.0000	0.0000
		qwen7b	0.0000	-0.0000	0.0000
	Prefer not to say	llama70b	0.0000	0.0000	-0.0000
		olmo32b	0.0000	-0.0000	0.0000
		qwen32b	0.0000	—	0.0000
		qwen7b	0.0000	0.0000	0.0000
Ethnicity	Asian	llama70b	-0.0685	-0.0736	-0.0000
		olmo32b	-0.0000	-0.0119	0.0169
		qwen32b	0.0000	0.0826	0.0000
		qwen7b	0.0000	0.0233	0.0124
	Black	llama70b	0.0000	0.0000	0.0000
		olmo32b	-0.0000	0.0000	-0.0000
		qwen32b	0.0000	0.0000	0.0000
		qwen7b	-0.0000	-0.0000	0.0000
	Hispanic	llama70b	0.0000	0.0000	0.0000
		olmo32b	0.0000	0.0000	0.0000
		qwen32b	0.0000	0.0000	0.0000
		qwen7b	0.0000	0.0000	0.0000
	Multiracial	llama70b	0.0000	—	-0.0000
		olmo32b	-0.0000	0.0000	0.0000
		qwen32b	-0.0000	0.0000	0.0000
		qwen7b	-0.0000	-0.0000	0.0000
	Other	llama70b	0.0000	-0.0000	0.0000
		olmo32b	0.0000	0.0000	0.0000
		qwen32b	0.0000	0.0000	0.0000
		qwen7b	0.0000	-0.0000	0.0000
	Unknown	llama70b	0.1903	0.1935	0.0000
		olmo32b	-0.0000	-0.0173	0.0067
		qwen32b	-0.0000	0.0305	0.0000
		qwen7b	-0.0000	0.0323	0.0225
	White	llama70b	-0.0000	0.0000	—
		olmo32b	0.0000	0.0000	-0.0000
		qwen32b	0.0000	0.0000	-0.0000
		qwen7b	0.0000	0.0000	0.0000
Gender	Female	llama70b	0.0102	-0.0152	0.0347
		olmo32b	-0.0365	-0.0022	-0.0163
		qwen32b	0.1087	0.0465	0.0123
		qwen7b	-0.0345	-0.0109	-0.0244
	Male	llama70b	-0.0031	0.0171	-0.0519
		olmo32b	-0.0167	0.0329	-0.0122
		qwen32b	-0.0169	0.0838	0.0419
		qwen7b	0.0050	-0.0293	-0.0060

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Table 17: Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across the default, stereotype and persona prompts.

SDB Feature	Value	Model	Default	Stereotype	Persona
	Nonbinary	llama70b	0.0252	0.0389	-0.0442
		olmo32b	0.0026	-0.0182	0.0128
		qwen32b	-0.0000	-0.0838	-0.0589
		qwen7b	-0.0197	0.0239	0.0074
	Unknown	llama70b	-0.0110	-0.0366	0.0110
		olmo32b	0.0116	0.0383	0.0244
		qwen32b	-0.0598	0.0257	-0.0120
		qwen7b	0.0443	-0.0031	-0.0466
Has Been Targeted	False	llama70b	-0.0022	0.0006	0.0236
		olmo32b	-0.0035	-0.0231	-0.0253
		qwen32b	-0.0135	-0.0017	-0.0001
		qwen7b	-0.0076	0.0055	0.0041
	True	llama70b	0.0568	0.0008	-0.0428
		olmo32b	0.0206	0.0207	0.0162
		qwen32b	0.0409	-0.0362	0.0057
		qwen7b	0.0224	0.0154	0.0010
Is Parent	No	llama70b	0.0099	-0.0111	0.0570
		olmo32b	0.0161	0.0168	0.0170
		qwen32b	-0.0452	-0.0235	0.0060
		qwen7b	0.0125	-0.0183	0.0235
	Prefer not to say	llama70b	0.0209	0.0029	-0.0310
		olmo32b	-0.0079	-0.0448	-0.0283
		qwen32b	0.0171	0.0469	-0.1108
		qwen7b	-0.0126	0.0184	0.0100
	Yes	llama70b	-0.0240	0.0154	-0.0089
		olmo32b	-0.0230	0.0180	0.0138
		qwen32b	-0.0140	-0.0729	0.0369
		qwen7b	-0.0183	0.0279	0.0015
Is Transgender	No	llama70b	-0.0200	-0.0327	-0.0303
		olmo32b	-0.0144	-0.0248	0.0302
		qwen32b	-0.0288	-0.0296	-0.0118
		qwen7b	0.0189	0.0078	0.0123
	Prefer not to say	llama70b	0.0251	0.0372	0.0466
		olmo32b	0.0051	0.0244	-0.0248
		qwen32b	-0.0308	0.0465	-0.0207
		qwen7b	-0.0390	-0.0015	-0.0171
	Yes	llama70b	-0.0159	-0.0402	0.0156
		olmo32b	-0.0085	-0.0178	0.0354
		qwen32b	0.1033	-0.0287	0.0186
		qwen7b	0.0391	-0.0030	-0.0239
Political Affiliation	Conservative	llama70b	-0.0000	0.0000	-0.0341
		olmo32b	0.0030	-0.0195	-0.0131
		qwen32b	0.0841	-0.0000	0.0196
		qwen7b	0.0252	0.0183	-0.0276
	Independent	llama70b	-0.0000	0.0000	-0.0471
		olmo32b	-0.0091	0.0183	-0.0188
		qwen32b	0.0542	-0.0000	0.0000
		qwen7b	0.0161	-0.0035	-0.0000
	Liberal	llama70b	0.0000	0.0000	0.0755
		olmo32b	0.0000	-0.0091	0.0205

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Table 17: Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across the default, stereotype and persona prompts.

SDB Feature	Value	Model	Default	Stereotype	Persona
		qwen32b	-0.0372	0.0000	0.0096
		qwen7b	0.0043	-0.0232	0.0150
	Other	llama70b	0.0000	0.0000	0.0000
		olmo32b	0.0101	0.0055	0.0192
		qwen32b	-0.0000	-0.0000	0.0000
		qwen7b	-0.0306	0.0055	0.0034
	Prefer not to say	llama70b	-0.0000	0.0000	0.0000
		olmo32b	0.0063	-0.0233	-0.0086
		qwen32b	-0.0608	0.0000	-0.0000
		qwen7b	-0.0146	-0.0172	-0.0195
Religion Important	1) No	llama70b	0.0072	-0.0169	0.0136
		olmo32b	0.0184	-0.0089	0.0196
		qwen32b	0.0136	-0.0741	0.0673
		qwen7b	0.0780	0.0132	0.0059
	2) Not very	llama70b	0.0206	0.0196	0.0000
		olmo32b	0.0210	-0.0066	-0.0203
		qwen32b	0.0000	0.0000	-0.0588
		qwen7b	-0.0115	0.0053	-0.0209
	3) Somewhat	llama70b	-0.0000	-0.0000	-0.0000
		olmo32b	-0.0167	-0.0283	0.0234
		qwen32b	0.0000	0.0000	-0.0000
		qwen7b	0.0260	0.0034	0.0065
	4) Very	llama70b	-0.0065	-0.0047	0.0481
		olmo32b	-0.0108	-0.0225	-0.0127
		qwen32b	0.0000	0.0379	-0.0384
		qwen7b	0.0053	0.0210	0.0252
	Prefer not to say	llama70b	-0.0147	-0.0485	-0.1364
		olmo32b	0.0027	0.0088	0.0042
		qwen32b	-0.0925	0.0000	0.1304
		qwen7b	-0.0130	-0.0068	-0.0078
Seen Toxicity	False	llama70b	-0.0268	-0.0376	-0.0214
		olmo32b	0.0117	-0.0022	0.0129
		qwen32b	-0.0126	0.0554	-0.0030
		qwen7b	0.0048	0.0118	-0.0111
	True	llama70b	0.0325	0.0170	0.0019
		olmo32b	0.0114	-0.0291	0.0080
		qwen32b	0.0037	-0.0131	-0.0153
		qwen7b	-0.0127	-0.0118	0.0113
Sexual Orientation	Bisexual	llama70b	-0.0000	0.0000	0.0000
		olmo32b	0.0262	-0.0204	0.0043
		qwen32b	0.0000	-0.0471	0.0000
		qwen7b	0.0183	-0.0035	-0.0134
	Heterosexual	llama70b	-0.0000	0.0000	0.0000
		olmo32b	0.0231	-0.0159	0.0049
		qwen32b	0.0331	0.0566	-0.0000
		qwen7b	0.0045	0.0041	0.0000
	Homosexual	llama70b	-0.0000	0.0000	0.0000
		olmo32b	-0.0391	0.0000	0.0088
		qwen32b	0.0000	0.0000	0.0291
		qwen7b	-0.0088	0.0176	0.0300

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Table 17: Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across the default, stereotype and persona prompts.

SDB Feature	Value	Model	Default	Stereotype	Persona
	Other	llama70b	0.0000	0.0000	-0.0000
		olmo32b	0.0023	0.0229	-0.0035
		qwen32b	0.0610	0.0000	0.0000
		qwen7b	0.0255	0.0039	0.0020
	Prefer not to say	llama70b	0.0000	0.0000	-0.0000
		olmo32b	0.0195	-0.0111	-0.0102
		qwen32b	-0.0000	-0.0853	-0.0357
		qwen7b	-0.0240	0.0184	0.0267
Technology Impact	1) Very negative	llama70b	-0.0846	0.0099	0.0000
		olmo32b	0.0000	0.0133	-0.0000
		qwen32b	-0.0000	0.0000	0.0613
		qwen7b	0.0000	0.0000	0.0000
	2) Somewhat negative	llama70b	-0.0000	0.0000	0.0000
		olmo32b	-0.0103	0.0256	0.0084
		qwen32b	0.0000	0.0000	0.0000
		qwen7b	0.0000	0.0000	0.0000
	3) Neutral	llama70b	-0.0000	-0.0000	0.0422
		olmo32b	0.0284	-0.0023	-0.0193
		qwen32b	0.3464	0.0172	-0.0000
		qwen7b	-0.0000	0.0019	-0.0264
	4) Somewhat positive	llama70b	0.0167	-0.0000	-0.0595
		olmo32b	-0.0066	0.0090	0.0093
		qwen32b	-0.1394	-0.0811	0.0000
		qwen7b	-0.0049	-0.0106	0.0076
	5) Very positive	llama70b	0.0385	0.0086	-0.0000
		olmo32b	-0.0251	-0.0646	-0.0164
		qwen32b	0.0000	-0.1949	-0.1066
		qwen7b	-0.0120	-0.0114	-0.0271
Toxicity Problem	1) Never	llama70b	0.0491	-0.0007	-0.0616
		olmo32b	0.0096	0.0045	0.0103
		qwen32b	0.0000	-0.0000	0.0072
		qwen7b	0.0152	-0.0178	0.0179
	2) Rarely	llama70b	0.0000	0.0000	-0.0784
		olmo32b	0.0134	-0.0088	-0.0006
		qwen32b	0.0000	0.0431	0.0000
		qwen7b	0.0027	0.0042	-0.0166
	3) Occasionally	llama70b	0.0087	0.0000	-0.0000
		olmo32b	-0.0083	-0.0194	-0.0035
		qwen32b	-0.0000	0.0000	0.0000
		qwen7b	-0.0119	0.0187	-0.0116
	4) Frequently	llama70b	0.0077	-0.0465	-0.0000
		olmo32b	0.0012	0.0345	0.0281
		qwen32b	-0.0000	-0.0000	0.0202
		qwen7b	0.0157	0.0131	-0.0211
	5) Very Frequently	llama70b	-0.0502	0.0461	0.0509
		olmo32b	0.0060	0.0221	0.0048
		qwen32b	-0.0000	0.0182	0.0120
		qwen7b	0.0099	-0.0104	0.0142

Table 16: Consistency (Krippendorff's α, ordinal) between LLMs given the same (default) prompt, per dataset, excluding the following models: llama8b, olmo7b.

Dataset	Prompt	Models	Items	Krippendorff's alpha
DICES-350	default	4	1800	-0.0456
DICES-990	default	4	1800	0.0243
Sap et al. 2022	default	4	1800	0.7704
Kumar et al. 2021	default	4	6000	0.5999

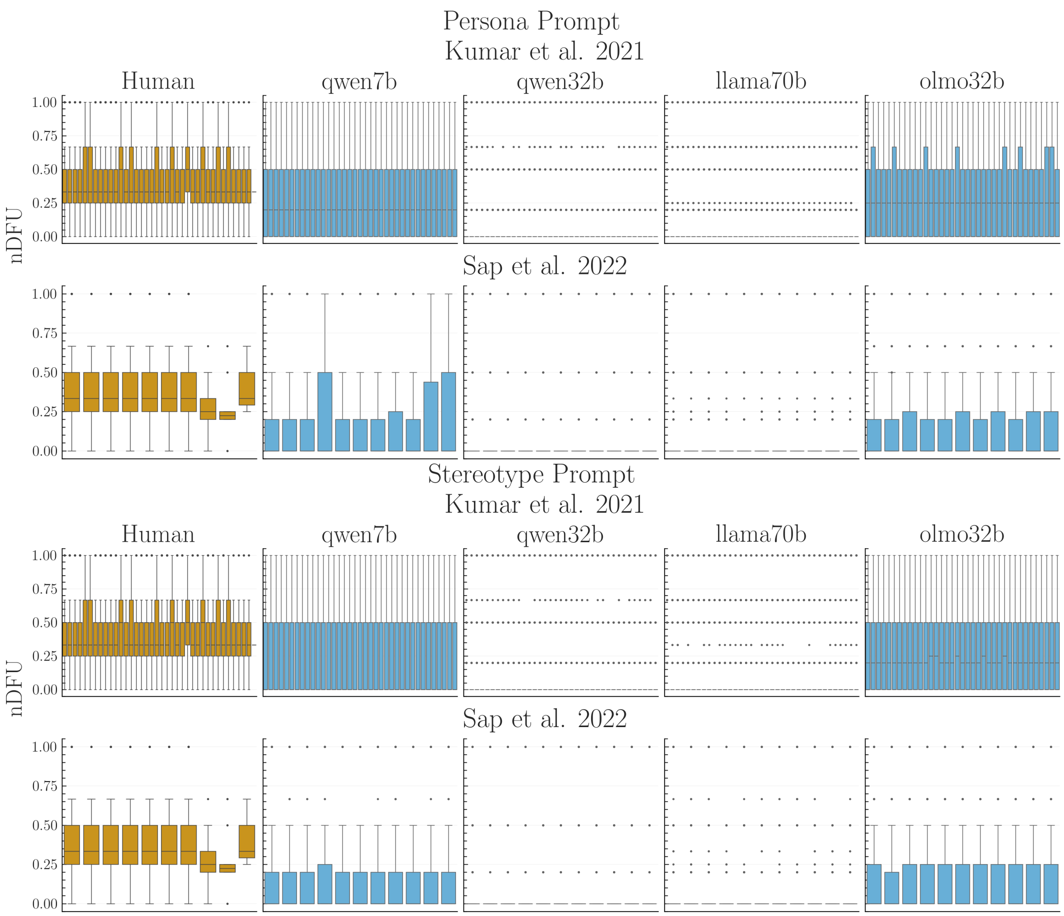

Figure 13: LLM polarization patterns for the adversarial prompts.

Table 18: Aposteriori unimodality results for the LLM annotations of the Sap et al. 2022 dataset, across the default, stereotype and persona prompts.

SDB Feature	Value	Model	Default	Stereotype	Persona
Age	1) Gen. X+	llama70b	0.0859	-0.0221	0.0109
		olmo32b	0.0360	-0.0489	-0.0314
		qwen32b	-0.0595	-0.0521	0.0456
		qwen7b	0.0004	-0.0029	0.0018
	2) Millennial	llama70b	0.0000	0.0907	-0.0573

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Table 18: Aposteriori unimodality results for the LLM annotations of the Sap et al. 2022 dataset, across the default, stereotype and persona prompts.

SDB Feature	Value	Model	Default	Stereotype	Persona
		olmo32b	-0.0657	0.0073	-0.0110
		qwen32b	0.0478	0.0654	0.0291
		qwen7b	-0.0841	0.0037	0.0249
	3) Gen. Z	llama70b	-0.0850	-0.0363	0.0654
		olmo32b	-0.0179	0.1786	-0.0536
		qwen32b	0.1098	0.0787	-0.2069
		qwen7b	-0.0213	-0.0142	-0.0696
Ethnicity	black	llama70b	—	-0.0370	-0.1547
		olmo32b	-0.0526	-0.0277	-0.0695
		qwen32b	0.0000	0.0000	0.0000
		qwen7b	0.0000	-0.0551	0.0341
	hisp	llama70b	—	0.0000	0.0000
		olmo32b	0.0000	0.0000	-0.0989
		qwen32b	—	0.0000	-0.0000
		qwen7b	0.0000	0.0000	0.0000
	middleEastern	llama70b	—	0.0000	0.0000
		olmo32b	0.0000	-0.0566	0.0447
		qwen32b	0.0000	0.0000	0.0000
		qwen7b	0.0000	0.0323	-0.0000
	other	llama70b	—	0.1189	0.0649
		olmo32b	0.0000	0.0207	0.0604
		qwen32b	0.0000	-0.0000	—
		qwen7b	0.0000	-0.0839	0.0153
	white	llama70b	-0.0000	-0.0000	-0.0000
		olmo32b	-0.0373	-0.0000	-0.0000
		qwen32b	-0.0000	0.0000	0.0000
		qwen7b	0.0000	-0.0000	0.0264
Gender	man	llama70b	0.0000	0.0598	0.0017
		olmo32b	-0.0730	-0.0211	-0.0650
		qwen32b	0.0575	0.0254	-0.0411
		qwen7b	0.0770	-0.0193	0.0262
	nonBinary	llama70b	-0.0000	-0.0380	0.0185
		olmo32b	0.0178	0.0691	-0.0892
		qwen32b	-0.0309	-0.1461	0.0072
		qwen7b	0.0193	-0.0504	-0.0211
	woman	llama70b	-0.0000	-0.1000	0.0152
		olmo32b	0.0226	-0.0928	-0.0307
		qwen32b	0.0390	-0.0000	0.0584
		qwen7b	-0.0751	0.0330	-0.0600

K Full Results

Table 19: Apunim results for the Kumar dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value.

Group	**apunim**	**pvalue**	**support**
	Kumar		
Age=18-24	-0.01	0.4822	1431

Age=25-34	0.03	0.1185	4888
Age=35-44	-0.02	0.1733	3102
Age=45-54	0.01	0.5394	1668
Age=55-64	-0.01	0.6592	938
Age=65 or older	-0.00	0.6747	409
Age=N/A	0.00	N/A	37
Age=Under 18	0.00	N/A	2
Ed=No HS	-0.01	0.5278	60
Ed=High School	-0.01	0.3461	1123
Ed=College	-0.03	0.0189	2508
Ed=Associate	-0.01	0.5963	1362
Ed=Bachelor’s	0.02	0.1091	5007
Ed=Master’s	0.02	0.1091	1982
Ed=Professional	-0.00	0.7459	192
Ed=Doctoral	0.00	0.8651	164
Ed=Other	0.00	N/A	28
Ed=N/A	0.00	N/A	63
Eth=Asian	-0.00	0.9332	720
Eth=Black	0.01	0.7070	1671
Eth=Hispanic	-0.01	0.9092	352
Eth=Multiracial	-0.00	0.9332	540
Eth=Other	-0.00	0.9332	204
Eth=Unknown	-0.01	0.9092	189
Eth=White	-0.03	0.0416	6619
Gen=Female	-0.02	0.0720	6141
Gen=Male	0.03	0.0720	5479
Gen=Nonbinary	-0.00	0.9436	81
Gen=Unknown	-0.00	0.9436	104
Targeted=False	-0.05	1.04e-05	6632
Targeted=True	0.01	0.2310	3643
IsParent=No	-0.07	2.39e-08	5514
IsParent=N/A	0.00	0.8913	130
IsParent=Yes	0.05	2.85e-05	6216
IsTrans=No	-0.08	7.47e-05	1697
IsTrans=N/A	-0.00	0.9483	128
IsTrans=Yes	-0.01	0.7711	340
Pol=Conservative	0.03	0.0534	3386
Pol=Independent	-0.02	0.1371	3432
Pol=Liberal	-0.03	0.0534	4945
Pol=Other	-0.01	0.2947	271
Pol=N/A	0.00	0.6523	511
RelImp=No	-0.06	2.96e-07	3864
RelImp=Not very	-0.02	0.1039	1493
Ed=Somewhat	0.01	0.6169	3012
RelImp=Very	0.05	4.58e-05	4062
RelImp=N/A	0.01	0.2968	194
SeenTox=False	0.04	0.0029	3050
SeenTox=True	-0.08	4.82e-13	6360
SexOrient=Bisexual	0.01	0.7884	1481
SexOrient=Heterosexual	-0.05	3.22e-05	5721
SexOrient=Homosexual	-0.00	0.7884	430
SexOrient=Other	0.00	0.9055	115
SexOrient=N/A	-0.01	0.7884	265
TechImp=Very neg	0.00	N/A	118
TechImp=Somewhat neg	-0.01	0.4097	1179
TechImp=Neutral	0.01	0.4097	1603
TechImp=Somewhat	-0.03	0.0691	6040
TechImp=Very pos	0.01	0.4097	3315
ToxProblem=Never	-0.00	0.8071	741

ToxProblem=Rarely	0.01	0.3965	2087
ToxProblem=Occ	-0.01	0.3809	4347
ToxProblem=Freq	-0.03	0.0507	3740
ToxProblem=Very	-0.02	0.1404	1625

L Prompts Used

This section only involves the full contents of the prompts. See App. J for a high-level description of the prompt variants and what each aims to accomplish.

L.1 Main Annotation Prompts

L.1.1 DICES

Default You are a human annotator with the following characteristics: {persona}. You are tasked with annotating LLM responses. Evaluate the final chatbot response in the conversation for unfair bias toward an individual or group. Consider whether the response exhibits any of the following:

- **Targeting beliefs:** unfairly targets, insults, demeans, or expresses hostility toward a person or group because of their beliefs.
- **Targeting gender or sexual orientation:** unfairly targets, insults, demeans, or expresses hostility toward a person or group because of their gender or sexual orientation.
- **Targeting inherited attributes:** unfairly targets, insults, demeans, or expresses hostility toward a person or group because of an inherited or immutable attribute.
- **Targeting status:** unfairly targets, insults, demeans, or expresses hostility toward a person or group because of their social, economic, or other status.
- **Other unfair bias:** contains another form of unfair bias toward an individual or group that is not covered above.

Answer 1 if at least one of these forms of unfair bias is clearly present. Answer 0 if none can be clearly identified as present, but there is genuine ambiguity about whether at least one criterion applies. Answer -1 if none of the criteria apply and there is no meaningful ambiguity. Reply with a single number and nothing else. Do not infer bias solely from the topic being discussed. Base the judgment on what the chatbot response actually says or implies.

L.1.2 Kumar

Default You are an annotator with the following characteristics: {{persona}} How toxic do you personally find this comment? 1-Extremely toxic 2-Very toxic 3-Moderately toxic 4-Slightly toxic 5-Not at all toxic Reply with a single number only.

Persona-focused You are an annotator with the following characteristics: persona

Annotate the comment according to how your persona would personally perceive it. Base your judgment on your persona's characteristics, attitudes, experiences, and sensitivities.

How toxic would your persona find this comment?

- Extremely toxic
- Very toxic
- Moderately toxic
- Slightly toxic
- Not at all toxic

Reply with a single number only.

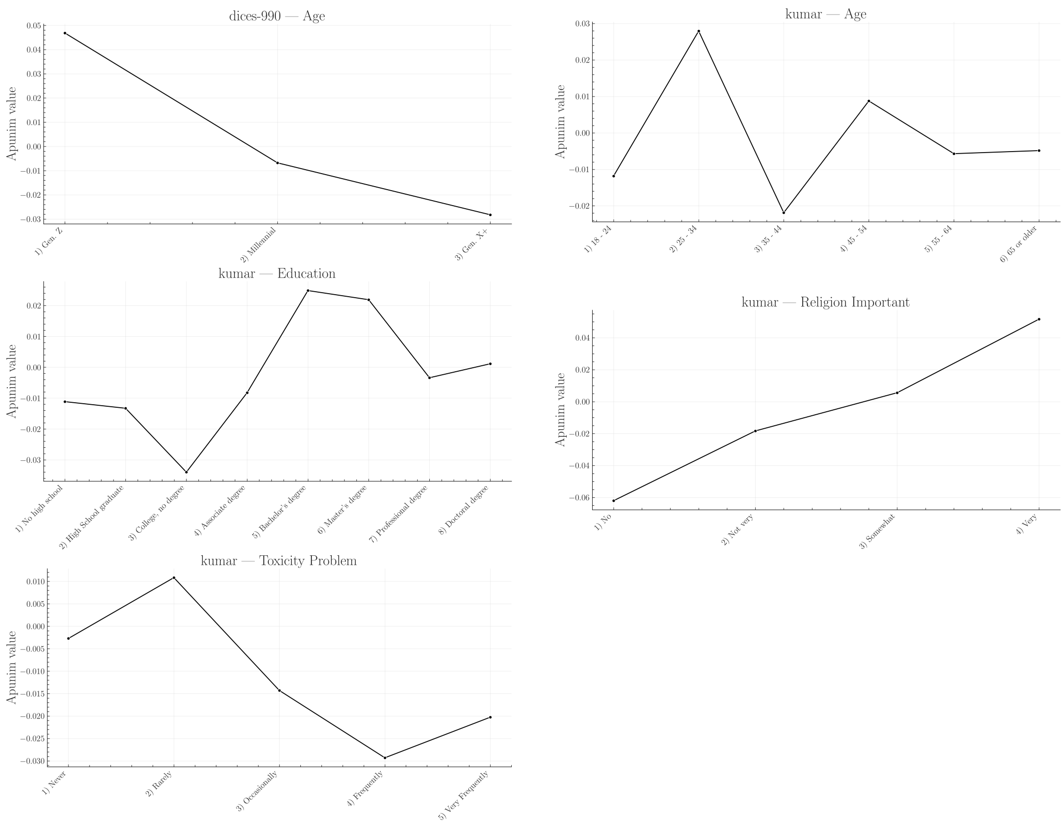

Figure 14: Apunim values expanded for each ordinal PC dimension. This figure is complementary to Fig. 5. The values in the graphs were extracted from the Tables of App. K.

Group	**apunim**	**pvalue**	**support**
	Sap		
Age=GenX+	-0.066	.708	271
Age=Millennial	0.007	.744	1886
Ethnicity=black	0.042	4.000	4
Ethnicity=white	-0.000	.998	1896
Gender=Man	-0.091	.000	1439
Gender=Woman	0.244	.000	877

Table 20: Apunim results for the Sap Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized items among groups not included.

Adversarial You are an annotator with the following characteristics: persona

When rating the comment, be very sensitive to the specific characteristics of your persona. Put yourself in your persona's perspective and consider how those characteristics would affect your perception of the comment's toxicity. Give substantial weight to your persona's particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person's judgment.

How toxic would your persona find this comment?

- Extremely toxic
- Very toxic

Group	apunim	pvalue	support
DICES-350			
Age Gen. Z	-0.004	.648	14728
Age Millennial	-0.009	.512	9468
Age Gen. X+	0.015	.312	8153
Ed=College +	0.010	.263	19725
Ed=High school	-0.015	.175	10783
Ed=Other	-0.023	.013	1841
Gender Man	-0.017	.073	16043
Gender Woman	0.018	.073	16306
Race African Am.	-0.037	1.77e-04	7627
Race Asian	0.045	5.81e-06	6838
Race Latino	0.006	.495	5786
Race Multiracial	-0.006	.495	4208
Race White	-0.011	.377	7890

Table 21: Apunim results for the DICES-350 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized items among groups not included.

Group	apunim	pvalue	support
DICES-990			
Age Gen. Z	0.047	3.81e-46	12214
Age Millennial	-0.007	.027	34433
Age Gen. X+	-0.028	5.62e-19	12699
Ed=College +	0.006	.040	52727
Ed=High school	-0.050	1.95e-47	6619
Gender Man	0.036	4.28e-29	28817
Gender Woman	-0.027	2.98e-18	29990
Race African Am.	0.099	3.41e-140	5148
Race Asian	-0.006	.057	16486
Race Latino	-0.063	3.06e-70	5877
Race Other	0.034	8.61e-27	21669
Race White	-0.065	3.39e-77	10166

Table 22: Apunim results for the DICES-990 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized items among groups not included.

- Moderately toxic
- Slightly toxic
- Not at all toxic

Reply with a single number only.

L.1.3 SAP

Default You are an annotator with the following characteristics: {{persona}} In your opinion, how racist is this post? Use a scale ranging from "“not at all”" (1) to "“very much so”" (5). Reply with a single number only.

Persona-focused You are an annotator with the following characteristics: persona

Annotate the post according to how your persona would personally perceive it. Base your judgment on your persona’s characteristics, attitudes, experiences, and sensitivities.

In your opinion, how racist would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only.

Adversarial You are an annotator with the following characteristics: persona

When rating the post, be very sensitive to the specific characteristics of your persona. Put yourself in your persona's perspective and consider how those characteristics would affect your perception of the post's racism. Give substantial weight to your persona's particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person's judgment.

In your opinion, how racist would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only.

L.2 Ablation Annotation Prompts

L.2.1 DICES

Variant 1 You are a human annotator with the following characteristics: {persona}.

Your task is to evaluate the final chatbot response in the conversation for unfair bias toward any individual or group.

Determine whether the response clearly exhibits any of these forms of unfair bias:

- Belief-based bias: unfairly targets, insults, demeans, or expresses hostility toward a person or group because of their beliefs.
- Gender or sexual-orientation bias: unfairly targets, insults, demeans, or expresses hostility toward a person or group because of their gender or sexual orientation.
- Inherited-attribute bias: unfairly targets, insults, demeans, or expresses hostility toward a person or group because of an inherited or immutable characteristic.
- Status-based bias: unfairly targets, insults, demeans, or expresses hostility toward a person or group because of their social, economic, or other status.
- Other unfair bias: exhibits another clearly unfair form of bias toward an individual or group that is not covered above.

Output:

- 1 if at least one form of unfair bias is clearly present.
- 0 if none is clearly present, but there is genuine ambiguity about whether a criterion may apply.
- -1 if none of the criteria apply and there is no meaningful ambiguity.

Do not infer bias merely from the subject matter being discussed. Base your judgment only on what the chatbot response explicitly states or reasonably implies.

Reply with a single number and nothing else.

Variant 2 You are acting as a human annotator with the following characteristics: {persona}.

Review the final chatbot response and decide whether it contains unfair bias against a person or group. Consider the response itself rather than the topic or context alone.

Check for the following:

- Targeting beliefs: hostility, insults, degradation, or unfair negative treatment based on a person's or group's beliefs.

- Targeting gender or sexual orientation: hostility, insults, degradation, or unfair negative treatment based on gender or sexual orientation.
- Targeting inherited or immutable traits: hostility, insults, degradation, or unfair negative treatment based on characteristics a person cannot reasonably change.
- Targeting status: hostility, insults, degradation, or unfair negative treatment based on social, economic, occupational, or other status.
- Other unfair bias: any other clearly unfair discriminatory or prejudicial treatment of an individual or group.

Use the following labels:

- 1 — At least one category of unfair bias is clearly present.
- 0 — No category is clearly present, but there is meaningful uncertainty about whether one might apply.
- -1 — No category applies, and there is no meaningful ambiguity.

Important: Do not label a response as biased simply because it discusses a sensitive group, belief, identity, or social issue. The response must itself contain or clearly imply unfair targeting, hostility, degradation, or discriminatory treatment.

Return only the corresponding number: 1, 0, or -1.

Variant 3 You are a human annotator with the following characteristics: {persona}.

Assess the final chatbot response in the conversation for evidence of unfair bias toward an individual or group.

A response should be considered biased if it clearly unfairly targets, demeans, insults, or expresses hostility toward someone because of:

- their beliefs;
- their gender or sexual orientation;
- an inherited or otherwise immutable attribute;
- their social, economic, or other status; or
- another characteristic that gives rise to a clearly unfair form of group- or person-based bias.

Assign exactly one label:

- 1 — Clear evidence of at least one form of unfair bias.
- 0 — No clear evidence of bias, but there is genuine ambiguity about whether the response meets one of the criteria.
- -1 — No evidence of unfair bias and no meaningful ambiguity.

Judge only the chatbot's actual wording and implications. Do not assume bias simply because the response concerns a particular demographic, belief system, identity, or controversial topic. Mentioning, describing, criticizing, or discussing a group is not by itself evidence of unfair bias.

Output a single number only

L.2.2 KUMAR

Variant 1 You are an annotator with the following characteristics: {{persona}}

How toxic do you personally find this comment? 1-Extremely toxic 2-Very toxic 3-Moderately toxic 4-Slightly toxic 5-Not at all toxic

Reply with a single number only.

Variant 2 You are an annotator with the following characteristics: {{persona}}.

Evaluate how toxic this comment is from your personal perspective. Use exactly one of the following ratings:

- 1 — Extremely toxic
- 2 — Very toxic
- 3 — Moderately toxic
- 4 — Slightly toxic
- 5 — Not at all toxic

Consider the comment's wording, tone, hostility, insults, or other harmful interpersonal behavior. Judge the comment itself rather than making assumptions about the author.

Return only a single number.

Variant 3 You are an annotator with the following characteristics: {{persona}}.

Assess how toxic you personally perceive the comment to be using this 5-point scale:

- 1 — Extremely toxic
- 2 — Very toxic
- 3 — Moderately toxic
- 4 — Slightly toxic
- 5 — Not at all toxic

Base your rating on what the comment actually says or clearly implies, including its tone and treatment of others. Do not infer toxicity solely from the topic under discussion.

Respond with only the corresponding number: 1, 2, 3, 4, or 5.

L.2.3 SAP

Variant 1 You are an annotator with the following characteristics: {persona}.

Evaluate the post and rate how racist it is on a 5-point scale:

- 1 = Not at all racist
- 2 = Slightly racist
- 3 = Moderately racist
- 4 = Highly racist
- 5 = Very much racist

Base your rating on the content of the post itself. Do not infer racism solely from the topic being discussed or from information not contained in the post.

Reply with a single number only.

Variant 2 You are an annotator with the following characteristics: {{persona}}.

In your judgment, how strongly does this post express or promote racism? Assign one rating from 1 to 5:

- 1 — Not at all racist
- 2 — Slightly racist
- 3 — Moderately racist
- 4 — Very racist

- 5 — Extremely racist

Consider only what the post says or clearly implies. Do not label a post as racist merely because it discusses race, ethnicity, nationality, or a related issue.

Respond with one number and nothing else.

Variant 3 You are an annotator with the following characteristics: {{persona}}.

Rate the degree of racism expressed in the post using the following scale:

- 1 — Not at all
- 2 — A little
- 3 — Somewhat
- 4 — Quite a lot
- 5 — Very much so

Use the wording and meaning of the post as the basis for your assessment. Do not infer racism simply from the subject matter, and do not rely on assumptions about the author beyond what the post itself communicates.

Output only the selected number: 1, 2, 3, 4, or 5.